SAP ABAP
JANUARY KIIT-DU BATCH
SPRINT EVALUATION

ADITI DEWANGAN
6362031

PURCHASE ORDER

CASE-STUDY NUMBER 18

PACKAGE USED:
Z6362031_PACKAGE

PROBLEM STATEMENT

In **Purchase Order management**, there is a lack of a **unified view** to track purchase orders at both header and item levels, making it difficult to **monitor procurement** activities, ensure **data accuracy**, and analyze **purchasing operations** efficiently.

Through this case study, we address the problem by creating a **unified solution** that integrates Purchase Order header and item data, providing a structured and **consolidated view** for efficient tracking, accurate data handling, and better analysis of **procurement operations**.

LIST OF CONTENTS

01 TABLE CREATION USING TMG

02 TABLE OUTPUT

03 ALV OOPS

04 ADOBE FORMS

05 ODATA SERVICES

06 CALLING BAPI

07 CDS VIEW AND EXTENSION VIEW

08 RAP READ-ONLY

09 RAP CRUD MANAGED

10 RAP CRUD UNMANAGED

11 RAP MATERIAL CREATION BAPI

12 RAP SENDING MAIL

13 RAP SINGLETON SM30

14 PROXY

PROCESS FLOW

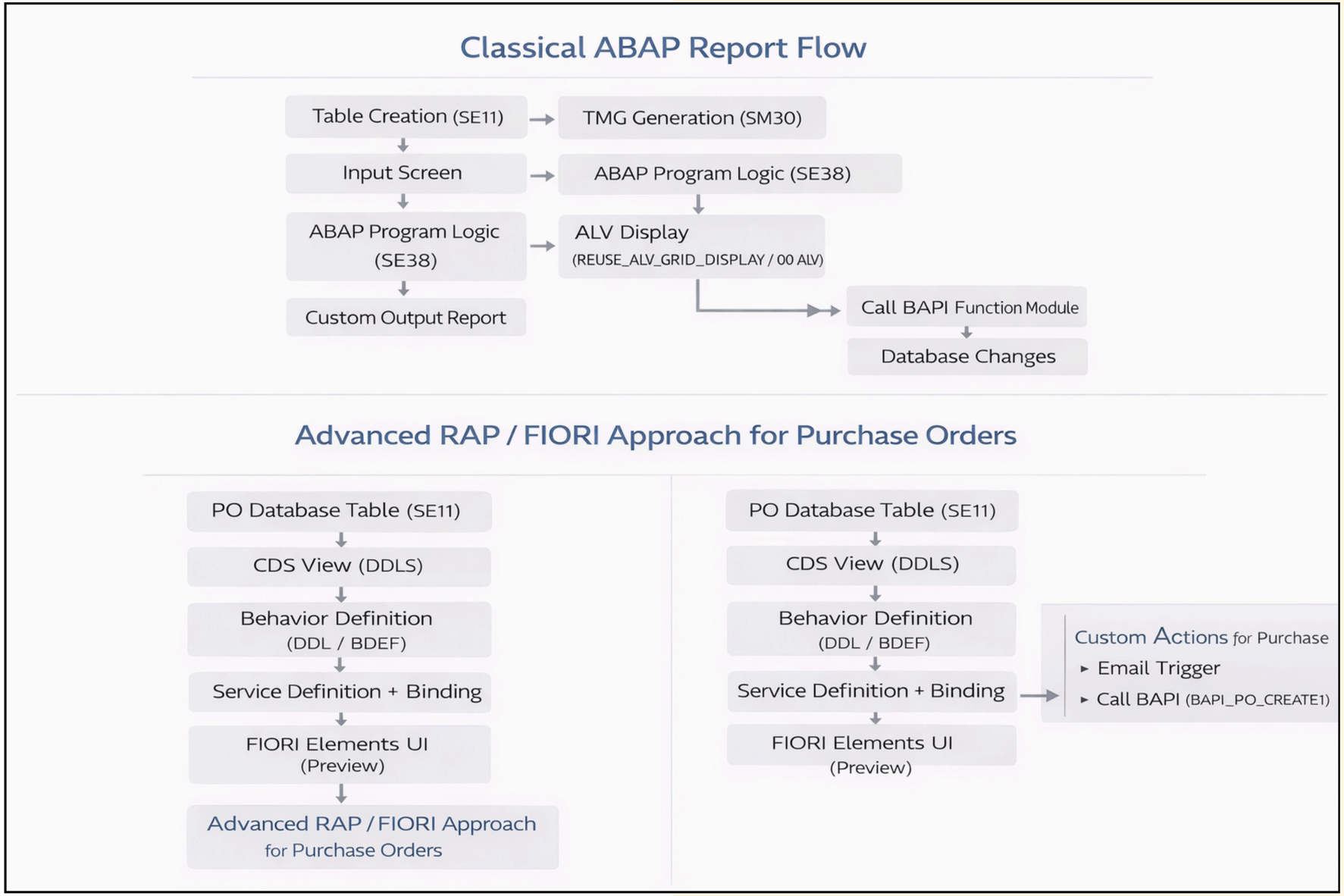


TABLE ZPO_EKKO__2031

Transparent Table: ZPO_EKKO__2031 Active

Short Description:* Purchasing Document Header

Attributes | Delivery and Maintenance | Fields | Input Help/Check | Currency/Quantity Fields | Indexes

Srch Help | Built-In Type

Field	Key	Initia...	Data element	Data T...	Length	De...	Coordi...	Short Description
MANDT	✓	✓	MANDT	CLNT	3	0	0	Client
EBELN	✓	✓	Z2031_EBELN	CHAR	10	0	0	Purchasing Document Number
BUKRS			BUKRS	CHAR	4	0	0	Company Code
BSTYP			EBSTYP	CHAR	1	0	0	Purchasing Document Category
BSART			ESART	CHAR	4	0	0	Purchasing Document Type
LIFNR			Z2031_LIFNR	CHAR	10	0	0	Supplier's Account Number
ZTERM			DZTERM	CHAR	4	0	0	Terms of Payment Key
EKORG			EKORG	CHAR	4	0	0	Purchasing Organization
EKGRP			BKGRP	CHAR	3	0	0	Purchasing Group
WAERS			WAERS	CUKY	5	0	0	Currency Key

ZPO_EKKO__2031

Fields: 10 of 10 Fixed Columns: 2 Lis

EBELN	BUKRS	BSTYP	BSART	LIFNR	ZTERM	EKORG	EKGRP	WAERS
3000000008	3000	F	EC	2	ZB01	3000	12	USD
3000000009	4000	F	EC	8	1	3000	10	USD
3000000010	4000	F	EC	8	1	3000	10	USD
3000000012	5000	F	EC	8	ZB01	3000	13	USD
3000000014	6000	F	EC	15	ZB01	3000	11	USD
3000000015	4000	F	EC	8	0	3000	11	USD
3000000016	7000	F	EC	100	ZB01	3000	11	USD
3000000018	3000	F	EC	100	ZB01	3000	12	USD

TABLE ZPO_EKPO_2031

Transparent Table: ZPO_EKPO_2031 Active

Short Description:* Purchasing Document Item

Attributes | Delivery and Maintenance | Fields | Input Help/Check | Currency/Quantity Fields | Indexes

Srch Help | Built-In Type

Field	Key	Initia...	Data element	Data Type	Length	Decimal...	Coor...	Short Description
MANDT	☑	☑	MANDT	CLNT	3	0	0	Client
EBELN	☑	☑	Z2031_EBELN	CHAR	10	0	0	Purchasing Document Number
EBELP	☑	☑	Z2031_EBELP	NUMC	5	0	0	Item Number of Purchasing Document
TXZ01	☐	☐	TXZ01	CHAR	40	0	0	Short Text
WERKS	☐	☐	EWERK	CHAR	4	0	0	Plant
MENGE	☐	☐	BSTMG	QUAN	13	3	0	Purchase Order Quantity
MEINS	☐	☐	BSTME	UNIT	3	0	0	Purchase Order Unit of Measure
NETWR	☐	☐	BWERT	CURR	13	2	0	Net Order Value in PO Currency
WEBAZ	☐	☐	WEBAZ	DEC	3	0	0	Goods receipt processing time in days
MWSKZ	☐	☐	MWSKZ	CHAR	2	0	0	Tax on sales/purchases code

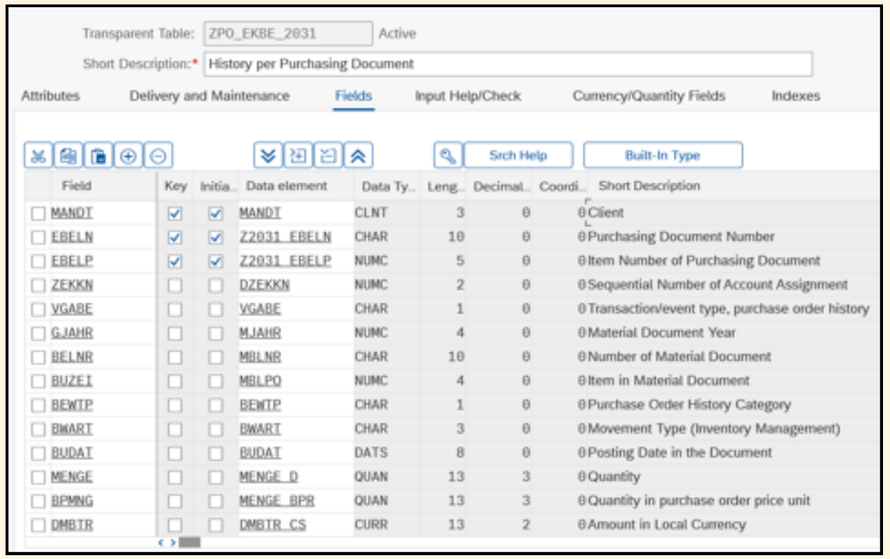
»»»

able: ZPO_EKPO_2031

Displayed Fields: 10 of 10 Fixed Columns: 3 List Width 0250

MANDT	EBELN	EBELP	TXZ01	WERKS	MENGE	MEINS	NETWR	WEBAZ	MWSKZ
100	3000000008	00001	Dishwasher Gel, Automatic	3200	2.000	ST	13.90	0	I0
100	3000000008	00002	Cleaner, Clear Vision Glass	3200	3.000	ST	29.85	0	I0
100	3000000008	00003	Cleaner, Concentrated Wood	3200	5.000	ST	99.75	0	I0
100	3000000008	00004	Cleaner, Tile & Grout	3200	4.000	ST	23.80	0	I0
100	3000000009	00001	Shafting assembly	3200	3.000	ST	360.00	0	I0
100	3000000009	00003	Pressure cover	3200	2.000	ST	324.00	0	I0
100	3000000009	00004	Rubber Seal	3200	7.000	ST	7.00	0	I0
100	3000000012	00003	Desk Pads	3200	3.000	ST	83.85	0	I0
100	3000000012	00004	Mouse Pads	3200	3.000	ST	32.85	0	I0
100	3000000012	00005	Pencil, Marking, Refill Black	3200	100.000	ST	104.00	0	I0
100	3000000014	00001	High Speed Printer/Laser Printer	3200	1.000	ST	2,228.00	3	I0
100	3000000014	00002	Maxitec R 3200 personal computer	3200	1.000	ST	1,698.00	0	I0
100	3000000015	00003	Mouse Pads	3200	1.000	ST	10.95	0	I0
100	3000000016	00002	High Speed Printer/Laser Printer	3200	1.000	ST	2,228.00	3	I0
100	3000000016	00003	Maxitec-R 3300 Professional PC	3200	1.000	ST	1,898.00	0	I0
100	3000000016	00004	PAQ Monitor, 21", Color	3200	1.000	ST	532.00	2	I0
100	3000000018	00001	Cleaner, Clear Vision Glass	3200	10.000	ST	99.50	0	I0
100	3000000018	00002	Cleaner, Concentrated Wood	3200	10.000	ST	199.50	0	I0

TABLE ZPO_EKBE_2031



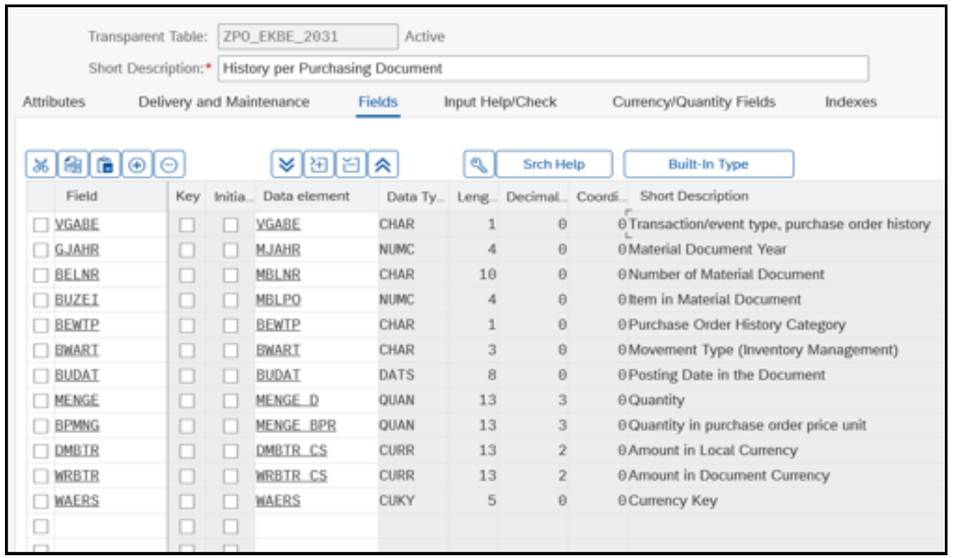
Transparent Table: ZPO_EKBE_2031 Active

Short Description:* History per Purchasing Document

Attributes | Delivery and Maintenance | Fields | Input Help/Check | Currency/Quantity Fields | Indexes

Srch Help | Built-In Type

Field	Key	Initia...	Data element	Data Ty...	Leng...	Decimal...	Coordi...	Short Description
MANDT	✓	✓	MANDT	CLNT	3	0	0	Client
EBELN	✓	✓	Z2031_EBELN	CHAR	10	0	0	Purchasing Document Number
EBELP	✓	✓	Z2031_EBELP	NUMC	5	0	0	Item Number of Purchasing Document
ZEKKN			DZEKKN	NUMC	2	0	0	Sequential Number of Account Assignment
VGABE			VGABE	CHAR	1	0	0	Transaction/event type, purchase order history
GJAHR			MJAHR	NUMC	4	0	0	Material Document Year
BELNR			MBLNR	CHAR	10	0	0	Number of Material Document
BUZEI			MBLPO	NUMC	4	0	0	Item in Material Document
BEWTP			BEWTP	CHAR	1	0	0	Purchase Order History Category
BWART			BWART	CHAR	3	0	0	Movement Type (Inventory Management)
BUDAT			BUDAT	DATS	8	0	0	Posting Date in the Document
MENGE			MENGE_D	QUAN	13	3	0	Quantity
BPMNG			MENGE_BPR	QUAN	13	3	0	Quantity in purchase order price unit
DMBTR			DMBTR_CS	CURR	13	2	0	Amount in Local Currency



Transparent Table: ZPO_EKBE_2031 Active

Short Description:* History per Purchasing Document

Attributes | Delivery and Maintenance | Fields | Input Help/Check | Currency/Quantity Fields | Indexes

Srch Help | Built-In Type

Field	Key	Initia...	Data element	Data Ty...	Leng...	Decimal...	Coordi...	Short Description
VGABE			VGABE	CHAR	1	0	0	Transaction/event type, purchase order history
GJAHR			MJAHR	NUMC	4	0	0	Material Document Year
BELNR			MBLNR	CHAR	10	0	0	Number of Material Document
BUZEI			MBLPO	NUMC	4	0	0	Item in Material Document
BEWTP			BEWTP	CHAR	1	0	0	Purchase Order History Category
BWART			BWART	CHAR	3	0	0	Movement Type (Inventory Management)
BUDAT			BUDAT	DATS	8	0	0	Posting Date in the Document
MENGE			MENGE_D	QUAN	13	3	0	Quantity
BPMNG			MENGE_BPR	QUAN	13	3	0	Quantity in purchase order price unit
DMBTR			DMBTR_CS	CURR	13	2	0	Amount in Local Currency
WRBTR			WRBTR_CS	CURR	13	2	0	Amount in Document Currency
WAERS			WAERS	CUKY	5	0	0	Currency Key

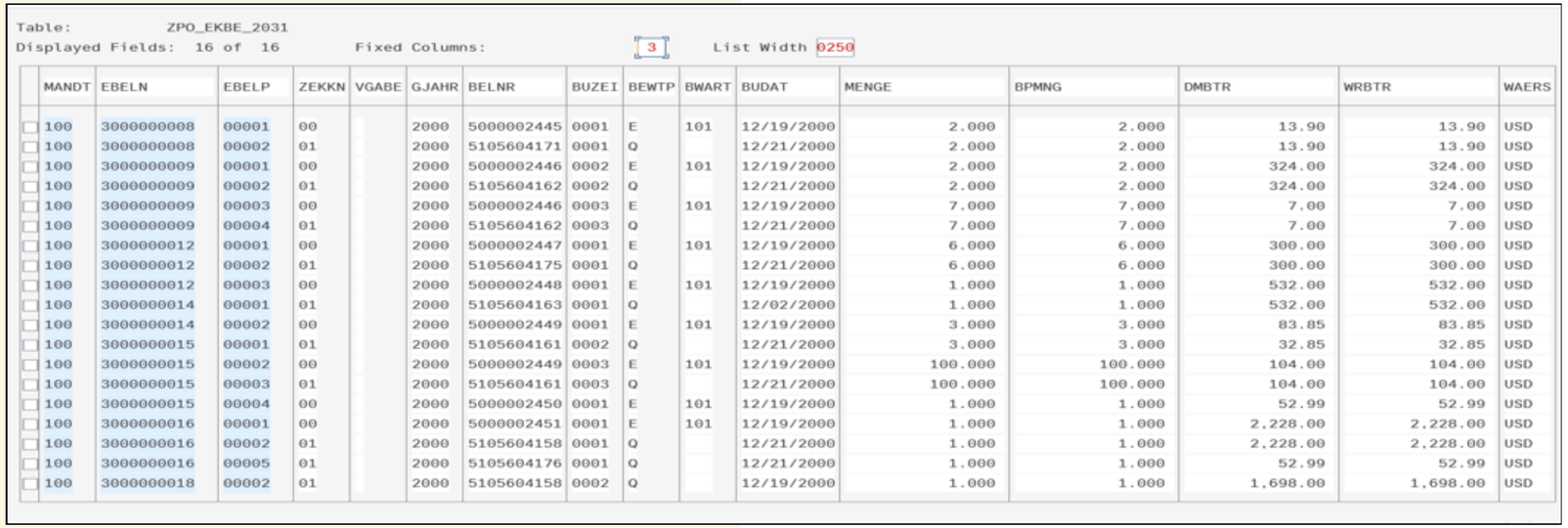
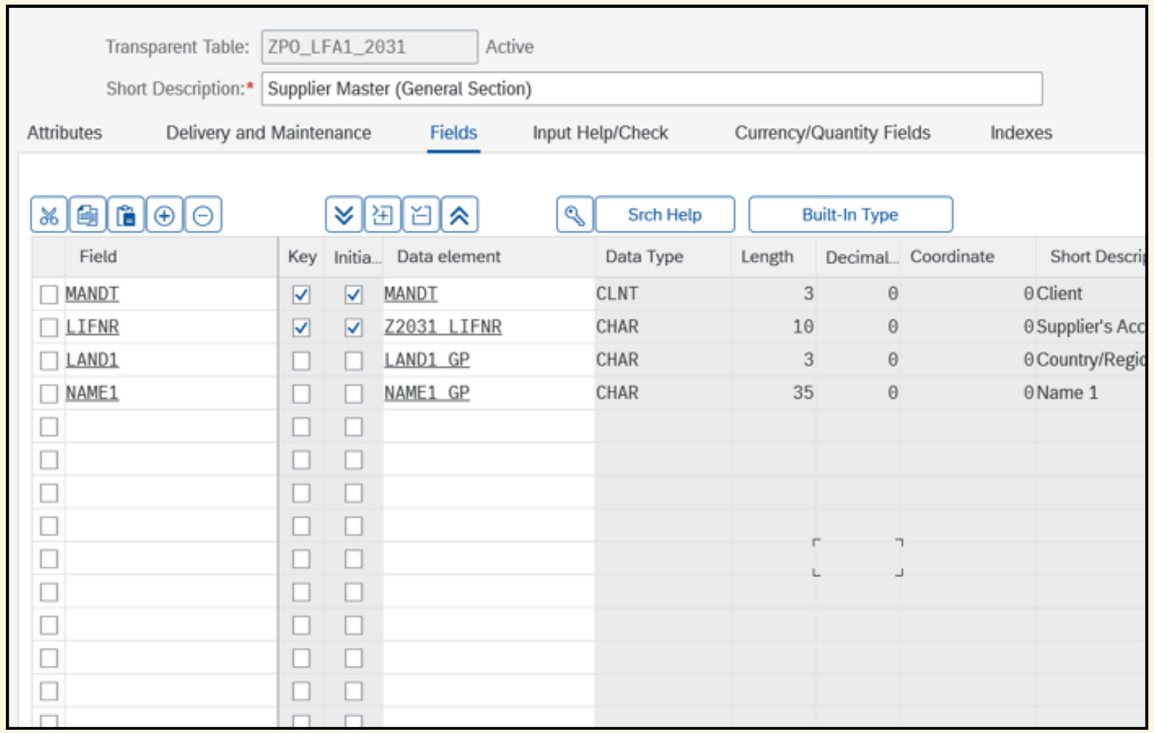


Table: ZPO_EKBE_2031

Displayed Fields: 16 of 16 Fixed Columns: 3 List Width 0250

MANDT	EBELN	EBELP	ZEKKN	VGABE	GJAHR	BELNR	BUZEI	BEWTP	BWART	BUDAT	MENGE	BPMNG	DMBTR	WRBTR	WAERS
100	3000000008	00001	00		2000	5000002445	0001	E	101	12/19/2000	2.000	2.000	13.90	13.90	USD
100	3000000008	00002	01		2000	5105604171	0001	Q		12/21/2000	2.000	2.000	13.90	13.90	USD
100	3000000009	00001	00		2000	5000002446	0002	E	101	12/19/2000	2.000	2.000	324.00	324.00	USD
100	3000000009	00002	01		2000	5105604162	0002	Q		12/21/2000	2.000	2.000	324.00	324.00	USD
100	3000000009	00003	00		2000	5000002446	0003	E	101	12/19/2000	7.000	7.000	7.00	7.00	USD
100	3000000009	00004	01		2000	5105604162	0003	Q		12/21/2000	7.000	7.000	7.00	7.00	USD
100	3000000012	00001	00		2000	5000002447	0001	E	101	12/19/2000	6.000	6.000	300.00	300.00	USD
100	3000000012	00002	01		2000	5105604175	0001	Q		12/21/2000	6.000	6.000	300.00	300.00	USD
100	3000000012	00003	00		2000	5000002448	0001	E	101	12/19/2000	1.000	1.000	532.00	532.00	USD
100	3000000014	00001	01		2000	5105604163	0001	Q		12/02/2000	1.000	1.000	532.00	532.00	USD
100	3000000014	00002	00		2000	5000002449	0001	E	101	12/19/2000	3.000	3.000	83.85	83.85	USD
100	3000000015	00001	01		2000	5105604161	0002	Q		12/21/2000	3.000	3.000	32.85	32.85	USD
100	3000000015	00002	00		2000	5000002449	0003	E	101	12/19/2000	100.000	100.000	104.00	104.00	USD
100	3000000015	00003	01		2000	5105604161	0003	Q		12/21/2000	100.000	100.000	104.00	104.00	USD
100	3000000015	00004	00		2000	5000002450	0001	E	101	12/19/2000	1.000	1.000	52.99	52.99	USD
100	3000000016	00001	00		2000	5000002451	0001	E	101	12/19/2000	1.000	1.000	2,228.00	2,228.00	USD
100	3000000016	00002	01		2000	5105604158	0001	Q		12/21/2000	1.000	1.000	2,228.00	2,228.00	USD
100	3000000016	00005	01		2000	5105604176	0001	Q		12/21/2000	1.000	1.000	52.99	52.99	USD
100	3000000018	00002	01		2000	5105604158	0002	Q		12/19/2000	1.000	1.000	1,698.00	1,698.00	USD

TABLE ZPO_LFAI_2031



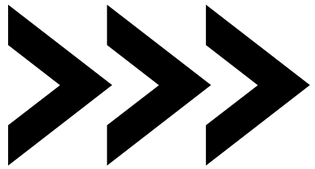
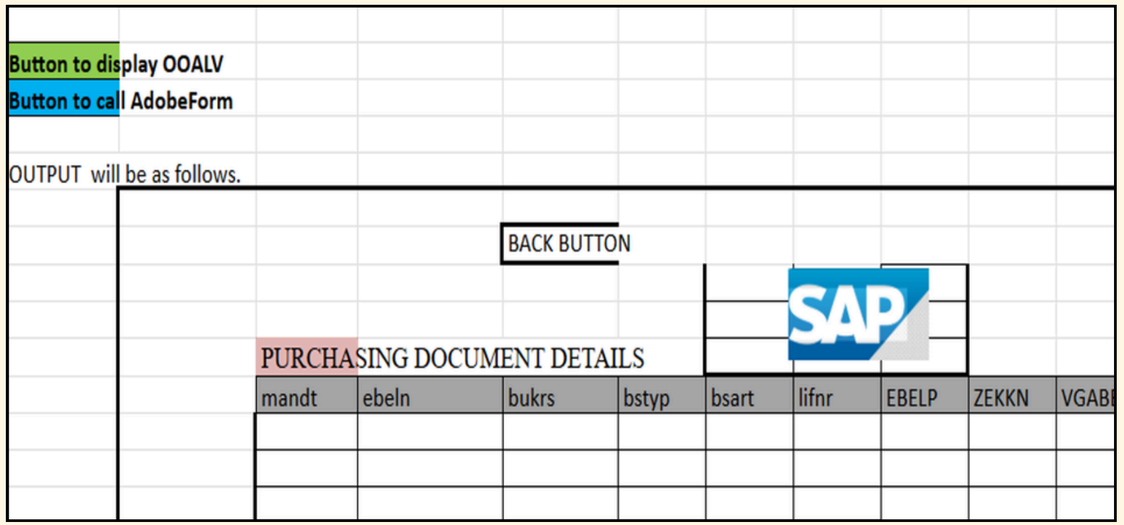


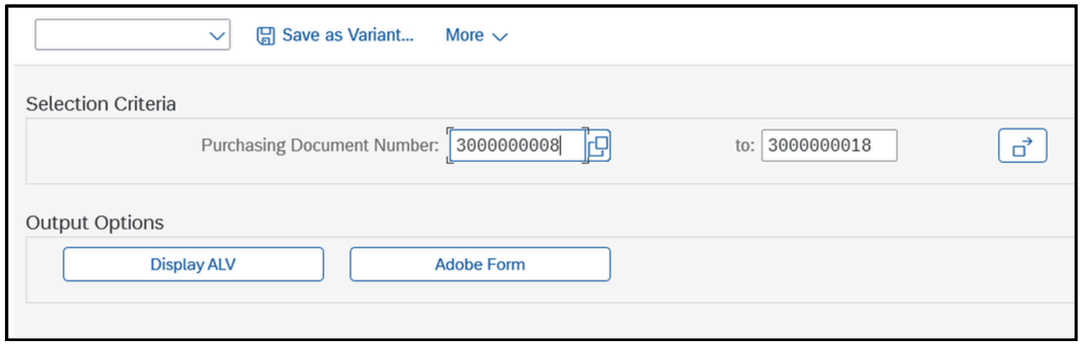
Table: ZPO_LFA1_2031
Displayed Fields: 4 of 4 Fixed Columns: 2 List

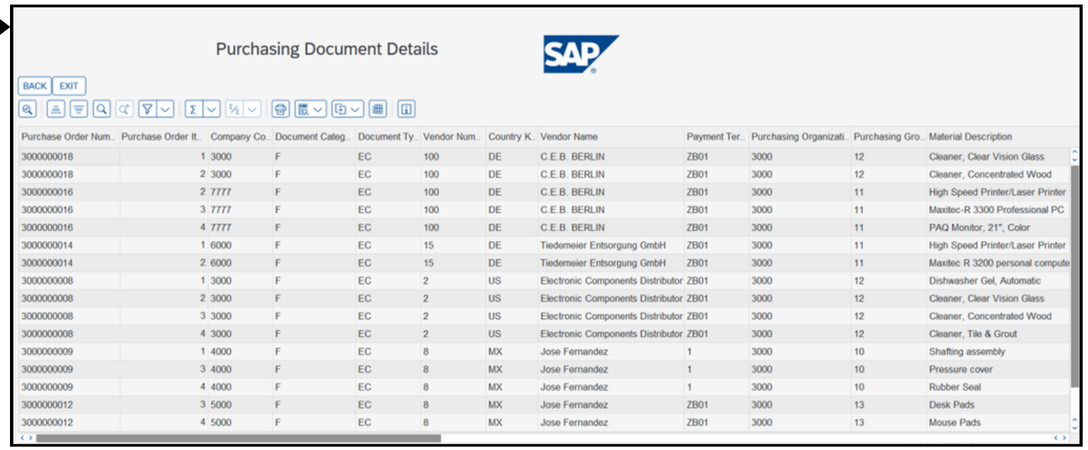
MANDT	LIFNR	LAND1	NAME1
100	1	DE	Forks Manufacturing GmbH
100	100	DE	C.E.B. BERLIN
100	111	DE	KBB Schwarze Pumpe
100	15	DE	Tiedemeier Entsorgung GmbH
100	2	US	Electronic Components Distributor
100	200	US	SMP
100	222	US	Express Vendor Inc
100	50	DE	Central Logistics Inc.
100	8	MX	Jose Fernandez

OOPS ALV EXPECTED AND ACTUAL OUTPUT

“Standard reports do not provide a structured and interactive way to display data, limiting efficient analysis and user interaction.”







Purchase Order Num..	Purchase Order It..	Company Co..	Document Categ..	Document Ty..	Vendor Num..	Country K..	Vendor Name	Payment Ter..	Purchasing Organizati..	Purchasing Gro..	Material Description
3000000018	1	3000	F	EC	100	DE	C.E.B. BERLIN	ZB01	3000	12	Cleaner, Clear Vision Glass
3000000018	2	3000	F	EC	100	DE	C.E.B. BERLIN	ZB01	3000	12	Cleaner, Concentrated Wood
3000000016	2	7777	F	EC	100	DE	C.E.B. BERLIN	ZB01	3000	11	High Speed Printer/Laser Printer
3000000016	3	7777	F	EC	100	DE	C.E.B. BERLIN	ZB01	3000	11	Maxitec-R 3300 Professional PC
3000000016	4	7777	F	EC	100	DE	C.E.B. BERLIN	ZB01	3000	11	PAQ Monitor, 21", Color
3000000014	1	6000	F	EC	15	DE	Tiedemeier Entsorgung GmbH	ZB01	3000	11	High Speed Printer/Laser Printer
3000000014	2	6000	F	EC	15	DE	Tiedemeier Entsorgung GmbH	ZB01	3000	11	Maxitec R 3200 personal compute
3000000008	1	3000	F	EC	2	US	Electronic Components Distributor	ZB01	3000	12	Dishwasher Gel, Automatic
3000000008	2	3000	F	EC	2	US	Electronic Components Distributor	ZB01	3000	12	Cleaner, Clear Vision Glass
3000000008	3	3000	F	EC	2	US	Electronic Components Distributor	ZB01	3000	12	Cleaner, Concentrated Wood
3000000008	4	3000	F	EC	2	US	Electronic Components Distributor	ZB01	3000	12	Cleaner, Tile & Grout
3000000009	1	4000	F	EC	8	MX	Jose Fernandez	1	3000	10	Shafting assembly
3000000009	3	4000	F	EC	8	MX	Jose Fernandez	1	3000	10	Pressure cover
3000000009	4	4000	F	EC	8	MX	Jose Fernandez	1	3000	10	Rubber Seal
3000000012	3	5000	F	EC	8	MX	Jose Fernandez	ZB01	3000	13	Desk Pads
3000000012	4	5000	F	EC	8	MX	Jose Fernandez	ZB01	3000	13	Mouse Pads

ADOBE FORMS OUTPUT

“Traditional outputs do not provide structured and professionally formatted documents, limiting effective presentation and printing of data.”

SAP Logo				Vendor Address(Display Fields from LFA1 Table)				
	Purchasing Document Category							
EKKO~EBELN	EKKO~BUKRS	EKKO~BSTYP	EKKO~BSART	EKKO~LIFNR	EKKO~ZTERM	EKKO~EKORG	EKKO~EKGRP	EKKO~WAERS

Purchase Header

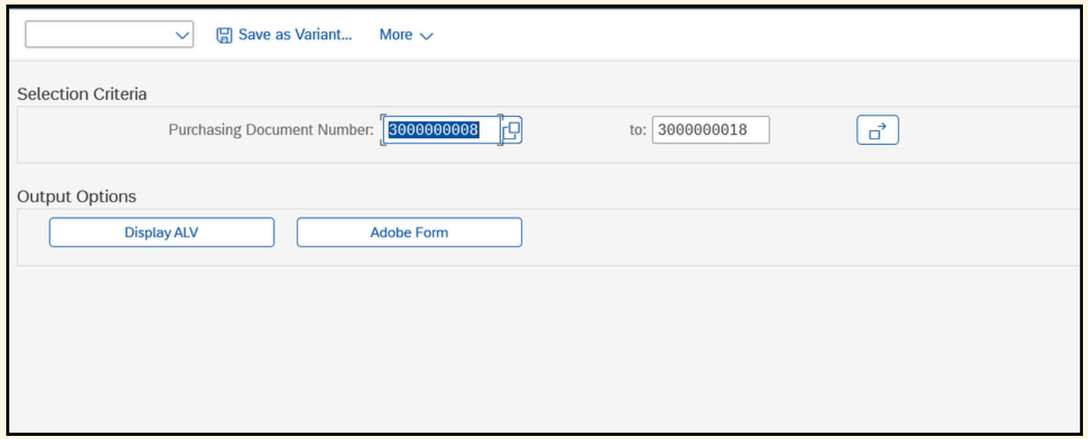
EKPO~EBELN	EKPO~EBELP	EKPO~TXZ01	EKPO~WEBAZ	EKPO~WERKS	EKPO~MWSKZ	EKPO~MENGE	EKPO~MEINS	EKPO~NETWR
EKBE~ZEKKN	EKBE~VGABE	EKBE~GJAHR	EKBE~BELNR	EKBE~BUZEI	EKBE~BEWTP	EKBE~BWART	EKBE~BUDAT	EKBE~MENGE
EKBE~BPMNG	EKBE~DMBTR	EKBE~WRBTR	EKBE~WAERS					

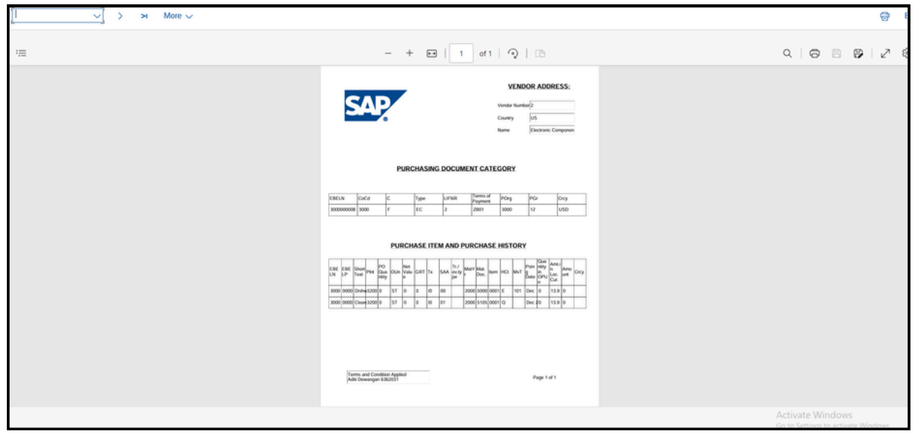
Purchase Item and Purchase History

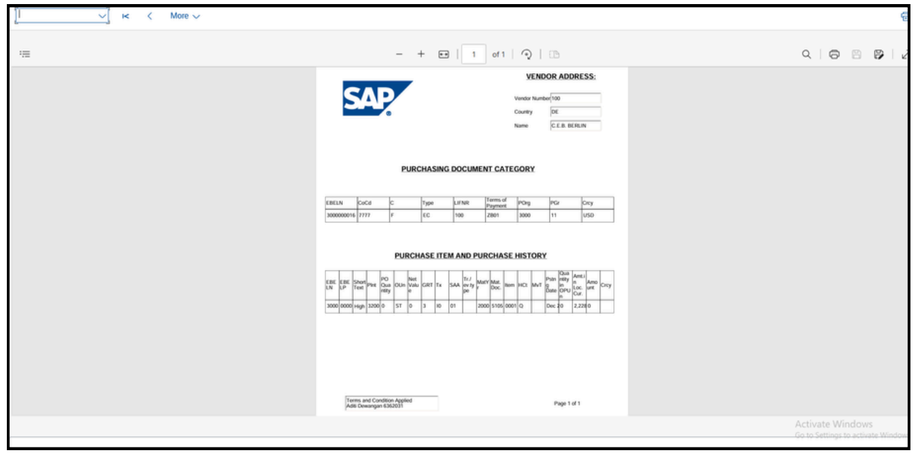
| | | | | | | | |

ADOBE FORMS OUTPUT

“Traditional outputs do not provide structured and professionally formatted documents, limiting effective presentation and printing of data.”







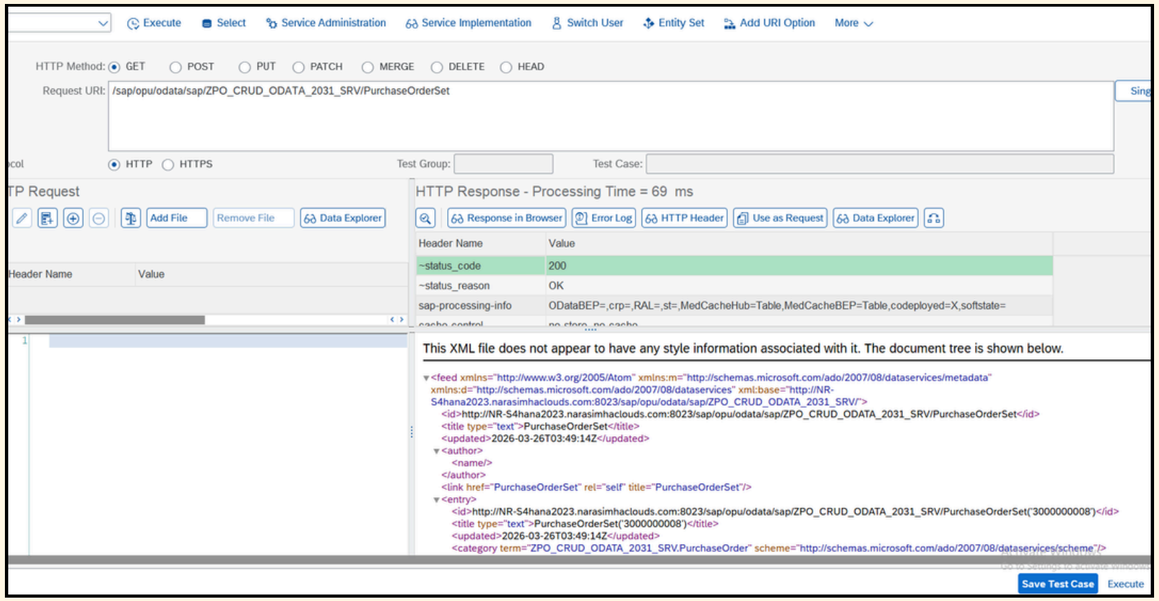
ODATA CRUD OPERATIONS

“Traditional systems lack a standardized way to expose and consume data, making integration between backend and UI applications difficult.”

Project

ZABAP24_NET2	Netweaver 2 practice without mapping
ZABAP24_NET3	Test Project
ZABAP24_NET4	kna1 crud operation
ZABAP24_NET5	vbak crud operations
ZABAP24_NET6_RAP	ODATA RAP
ZPO_CRUD_ODATA_2031	CRUD Operations in EKKO table
Data Model	
Entity Types	
PurchaseOrder	
Associations	
Entity Sets	
PurchaseOrderSet	
Service Implementation	
Runtime Artifacts	
ZCL_ZPO_CRUD_ODATA_203_DPC	
ZCL_ZPO_CRUD_ODATA_203_DPC_EXT	
ZCL_ZPO_CRUD_ODATA_203_MPC	
ZCL_ZPO_CRUD_ODATA_203_MPC_EXT	
ZPO_CRUD_ODATA_2031_MDL	

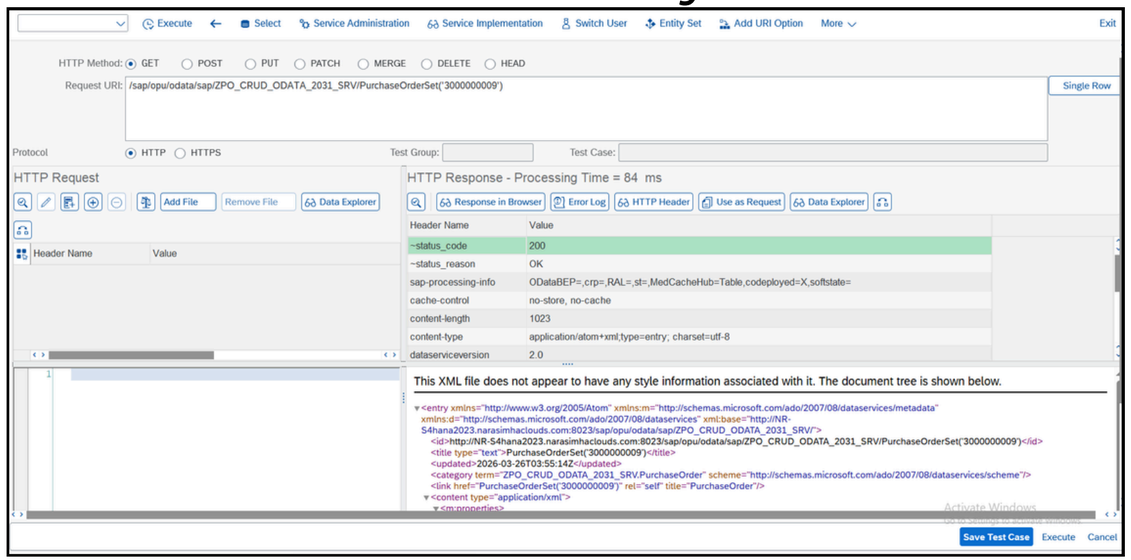
Get Entity_set



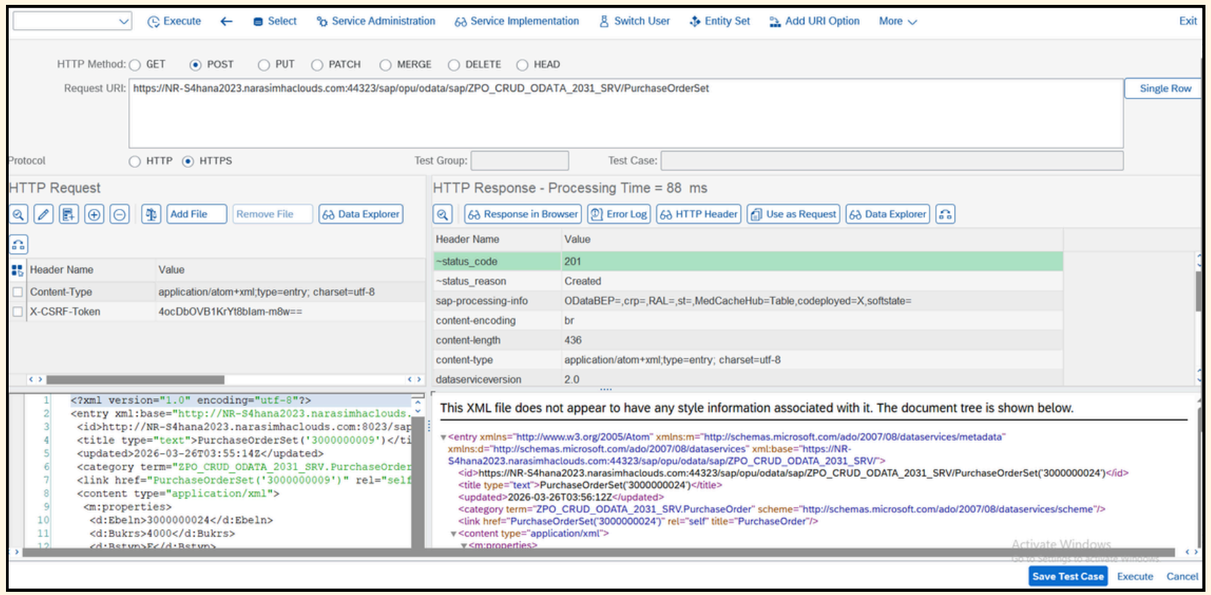
ODATA CRUD OPERATIONS

“OData provides a standardized service to expose and consume data with seamless integration between SAP backend and UI applications.”

Get Entity



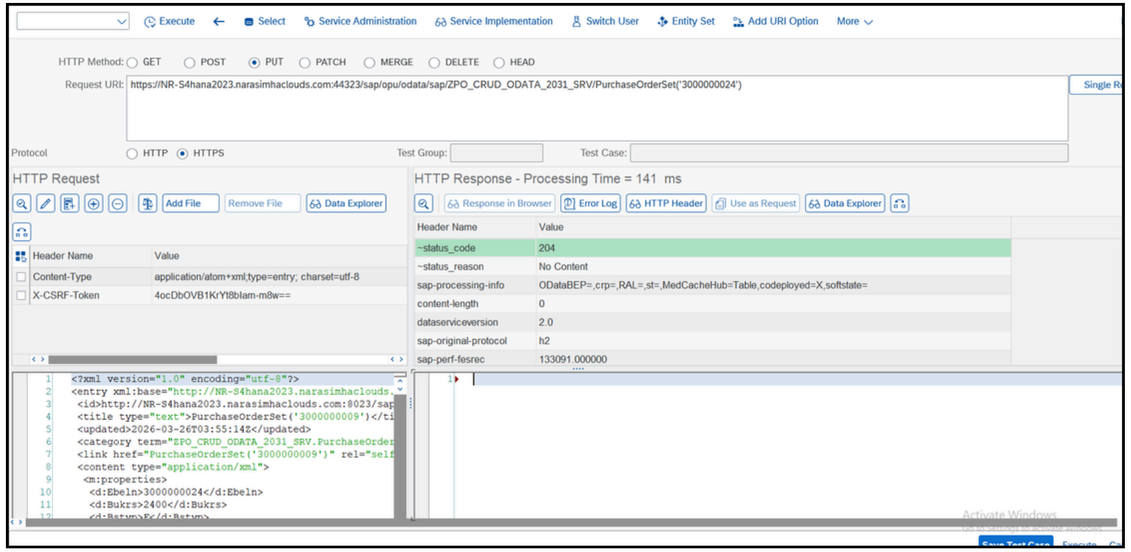
Create Entity



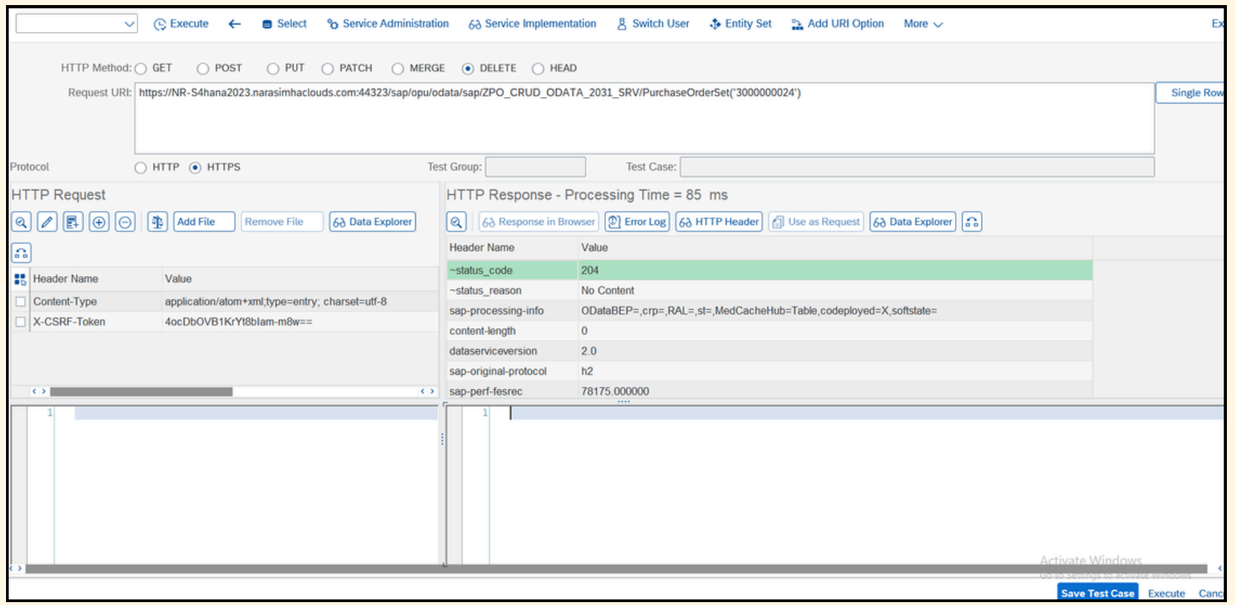
ODATA CRUD OPERATIONS

“OData provides a standardized service to expose and consume data with seamless integration between SAP backend and UI applications.”

Update Entity



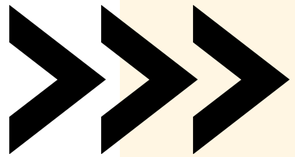
Delete Entity



BAPI FUNCTION CALL

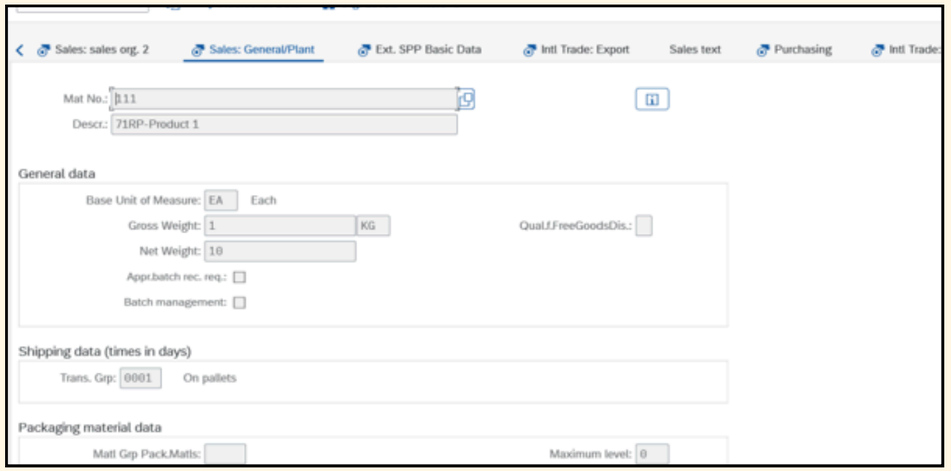
“Business operations lack a standardized and reliable method of execution, making data processing inconsistent and difficult to manage.”

```
21
22  * convert material to internal format
23  CALL FUNCTION 'CONVERSION_EXIT_MATN1_INPUT'
24    EXPORTING
25      input  = p_matnr
26    IMPORTING
27      output = lv_matnr.
28
29  * fetch material details
30  CALL FUNCTION 'BAPI_MATERIAL_GET_DETAIL'
31    EXPORTING
32      material = lv_matnr
33    IMPORTING
34      material_general_data = ls_clientdata
35      return                = ls_return_get.
36
37  * check if any error while fetching
38  IF ls_return_get-type = 'E' OR ls_return_get-type = 'A'.
39    WRITE: / 'Error:', ls_return_get-message.
40    EXIT.
41  ENDIF.
42
43  * updating net weight to 10
44  ls_clientdata_save-net_weight = 10.
45
46  * marking field for update
47  ls_clientdatax-net_weight = 'X'.
48
49  * header data for save
50  ls_head-material   = lv_matnr.
51  ls_head-basic_view = 'X'.
```



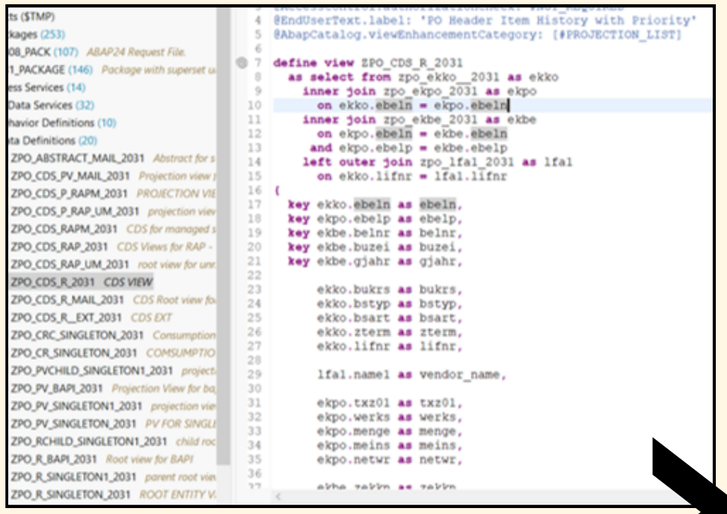



```
Report: ZPO_BAPI_REP_2031                    Active
48
49  * header data for save
50  ls_head-material   = lv_matnr.
51  ls_head-basic_view = 'X'.
52
53  * saving updated material
54  CALL FUNCTION 'BAPI_MATERIAL_SAVEDATA'
55    EXPORTING
56      headdata    = ls_head
57      clientdata  = ls_clientdata_save
58      clientdatax = ls_clientdatax
59    IMPORTING
60      return      = ls_return_save.
61
62  * commit to database
63  CALL FUNCTION 'BAPI_TRANSACTION_COMMIT'
64    EXPORTING
65      wait = 'X'.
66
67  * display result
68  IF ls_return_save-type = 'S'.
69    WRITE: / 'Success:', ls_return_save-message.
70  ELSE.
71    WRITE: / ls_return_save-type, ls_return_save-message.
72  ENDIF.
73
74  * final confirmation
75  WRITE: / 'Material updated successfully. Check MM03.'.
```



CDS VIEW AND ITS OUTPUT

“Traditional data models lack structure and optimization and reusability, making it difficult to efficiently integrate data with UI and services.”



```
@EndUserText.label: 'PO Header Item History with Priority'
@AbapCatalog.viewEnhancementCategory: [#PROJECTION_LIST]

define view ZPO_CDS_R_2031
  as select from zpo_ekko__2031 as ekko
    inner join zpo_ekpo_2031 as ekpo
      on ekko.ebeln = ekpo.ebeln
    inner join zpo_ekbe_2031 as ekbe
      on ekpo.ebeln = ekbe.ebeln
     and ekpo.ebelp = ekbe.ebelp
    left outer join zpo_lfa1_2031 as lfa1
      on ekko.lifnr = lfa1.lifnr
{
  key ekko.ebeln as ebeln,
  key ekpo.ebelp as ebelp,
  key ekbe.belnr as belnr,
  key ekbe.buzei as buzei,
  key ekbe.gjahr as gjahr,

      ekko.bukrs as bukrs,
      ekko.bstyp as bstyp,
      ekko.bsart as bsart,
      ekko.zterm as zterm,
      ekko.lifnr as lifnr,

      lfa1.name1 as vendor_name,

      ekpo.txz01 as txz01,
      ekpo.werks as werks,
      ekpo.menge as menge,
      ekpo.meins as meins,
      ekpo.netwr as netwr,
```



```
  key ekbe.gjahr as gjahr,

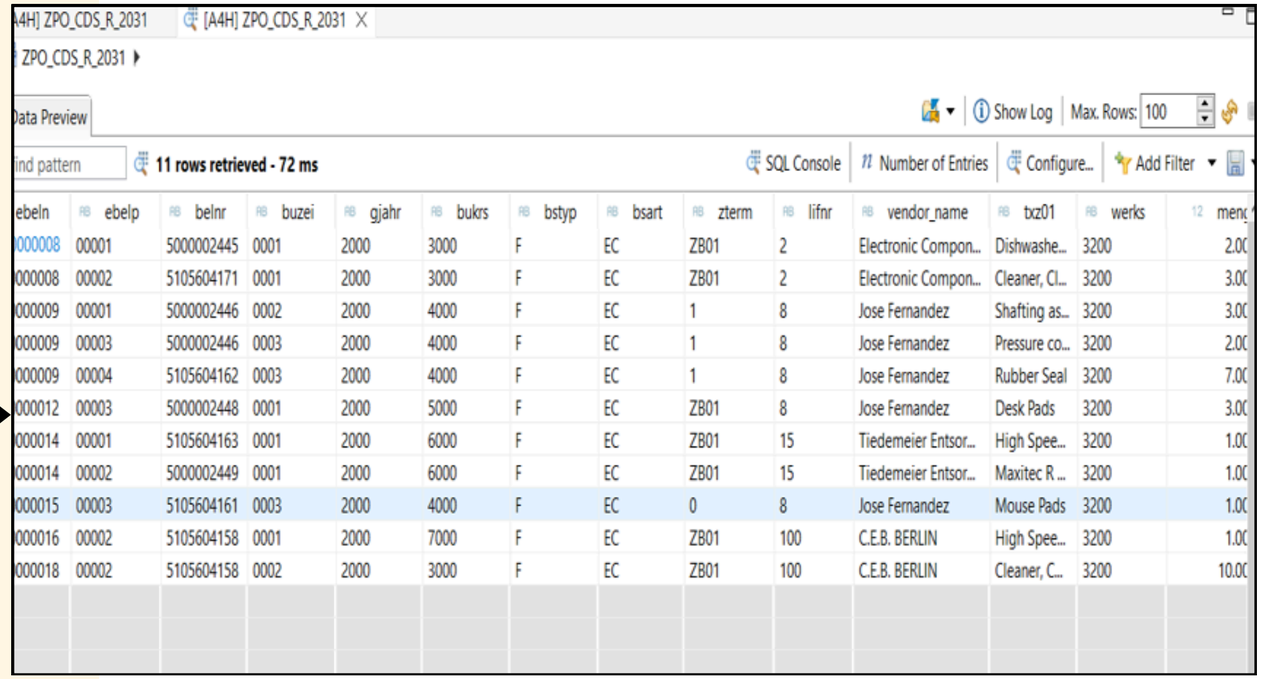
      ekko.bukrs as bukrs,
      ekko.bstyp as bstyp,
      ekko.bsart as bsart,
      ekko.zterm as zterm,
      ekko.lifnr as lifnr,

      lfa1.name1 as vendor_name,

      ekpo.txz01 as txz01,
      ekpo.werks as werks,
      ekpo.menge as menge,
      ekpo.meins as meins,
      ekpo.netwr as netwr,

      ekbe.zekkn as zekkn,
      ekbe.vgabe as vgabe,
      ekbe.bewtp as bewtp,

      cast(
        case
          when ekpo.netwr between 1 and 5000 then 'LOW      '
          when ekpo.netwr between 5001 and 10000 then 'MEDIUM
          when ekpo.netwr > 10000 then 'HIGH       '
          else 'UNKNOWN    '
        end
        as abap.char(10)
      ) as priority
}
```



[A4H] ZPO_CDS_R_2031 | [A4H] ZPO_CDS_R_2031

ZPO_CDS_R_2031

Data Preview | Show Log | Max. Rows: 100

Find pattern | 11 rows retrieved - 72 ms | SQL Console | Number of Entries | Configure... | Add Filter

ebeln	ebelp	belnr	buzei	gjahr	bukrs	bstyp	bsart	zterm	lifnr	vendor_name	txz01	werks	meng
000008	00001	5000002445	0001	2000	3000	F	EC	ZB01	2	Electronic Compon...	Dishwashe...	3200	2.00
000008	00002	5105604171	0001	2000	3000	F	EC	ZB01	2	Electronic Compon...	Cleaner, Cl...	3200	3.00
000009	00001	5000002446	0002	2000	4000	F	EC	1	8	Jose Fernandez	Shafting as...	3200	3.00
000009	00003	5000002446	0003	2000	4000	F	EC	1	8	Jose Fernandez	Pressure co...	3200	2.00
000009	00004	5105604162	0003	2000	4000	F	EC	1	8	Jose Fernandez	Rubber Seal	3200	7.00
000012	00003	5000002448	0001	2000	5000	F	EC	ZB01	8	Jose Fernandez	Desk Pads	3200	3.00
000014	00001	5105604163	0001	2000	6000	F	EC	ZB01	15	Tiedemeier Entsor...	High Spee...	3200	1.00
000014	00002	5000002449	0001	2000	6000	F	EC	ZB01	15	Tiedemeier Entsor...	Maxitec R ...	3200	1.00
000015	00003	5105604161	0003	2000	4000	F	EC	0	8	Jose Fernandez	Mouse Pads	3200	1.00
000016	00002	5105604158	0001	2000	7000	F	EC	ZB01	100	C.E.B. BERLIN	High Spee...	3200	1.00
000018	00002	5105604158	0002	2000	3000	F	EC	ZB01	100	C.E.B. BERLIN	Cleaner, C...	3200	10.00

CDS EXTENSION VIEW AND ITS OUTPUT

“Standard CDS views do not provide flexibility to add custom fields and logic without affecting the original definition.”

[A4H] ZPO_CDS_R_2031 | [A4H] ZPO_CDS_R_2031 | [A4H] ZPO_CDS_R_EXT_2031

```
@AbapCatalog.sqlViewAppendName: 'ZPOPRIOEXT'
@EndUserText.label: 'Extension for PO Priority CDS'

extend view ZPO_CDS_R_2031 with ZPO_CDS_R_EXT_2031
{
  ekko.ekorg as ekorg,
  ekko.ekgrp as ekgrp,
  ekko.waers as waers,

  ekpo.webaz as webaz,
  ekpo.mwskz as mwskz,

  ekbe.bwart as bwart,
  ekbe.budat as budat
}
```

[A4H] ZPO_CDS_R_2031 | [A4H] ZPO_CDS_R_2031 | [A4H] ZPO_CDS_R_EXT_2031 | [A4H] ZPO_CDS_R_EXT_2031

ZPO_CDS_R_2031

Data Preview — Show Log — Max. Rows: 100

find pattern — 11 rows retrieved - 5 ms — SQL Console — Number of Entries — Configure... — Add Filter

menge	meins	netwr	zekkn	vgabe	bewtp	priority	ekorg	ekgrp	waers	webaz	mwskz	bwart	budat
2.000	ST	13.90	00		E	LOW	3000	12	USD	0	I0	101	2000-12-19
3.000	ST	29.85	01		Q	LOW	3000	12	USD	0	I0		2000-12-21
3.000	ST	360.00	00		E	LOW	3000	10	USD	0	I0	101	2000-12-19
2.000	ST	324.00	00		E	LOW	3000	10	USD	0	I0	101	2000-12-19
7.000	ST	7.00	01		Q	LOW	3000	10	USD	0	I0		2000-12-21
3.000	ST	83.85	00		E	LOW	3000	13	USD	0	I0	101	2000-12-19
1.000	ST	2228.00	01		Q	LOW	3000	11	USD	3	I0		2000-12-02
1.000	ST	1698.00	00		E	LOW	3000	11	USD	0	I0	101	2000-12-19
1.000	ST	10.95	01		Q	LOW	3000	11	USD	0	I0		2000-12-21
1.000	ST	2228.00	01		Q	LOW	3000	11	USD	3	I0		2000-12-21
10.000	ST	199.50	01		Q	LOW	3000	12	USD	0	I0		2000-12-19

RAP READ-ONLY INPUT SCREEN

“There is a need to provide controlled access to data without allowing modifications to ensure data integrity.”

[A4H] ZPO_CDS_RAP_2031 | [A4H] ZPO_SD_RAP_2031 | [A4H] ZPO_SB_RAP_2031

```
@AccessControl.authorizationCheck: #NOT_REQUIRED
@EndUserText.label: 'PO Header Item History with Priority'
@Metadata.allowExtensions: true
@UI.headerInfo: {
  typeName: 'Purchase Order History',
  typeNamePlural: 'Purchase Order Histories',
  title: { value: 'Ebeln' },
  description: { value: 'Txz01' }
}
define root view entity ZPO_CDS_RAP_2031
  as select from zpo_ekko__2031 as a
    inner join zpo_ekpo_2031  as b
      on a.ebeln = b.ebeln
    inner join zpo_ekbe_2031  as c
      on b.ebeln = c.ebeln
     and b.ebelp = c.ebelp
    inner join zpo_lfa1_2031  as d
      on a.lifnr = d.lifnr
{
  @UI.facet: [ {
      id: 'GeneralInfo',
      type: #IDENTIFICATION_REFERENCE,
      label: 'General Information',
      position: 10
  } ]

  @UI.lineItem:       [ { position: 10,  label: 'PO Number' } ]
  @UI.identification: [ { position: 10,  label: 'PO Number' } ]
  key a.ebeln as Ebeln,

  @UI.lineItem:       [ { position: 20,  label: 'PO Item' } ]
  @UI.identification: [ { position: 20,  label: 'PO Item' } ]
  key b.ebelp as Ebelp,

  @UI.lineItem:       [ { position: 30,  label: 'Accounting Doc' } ]
  @UI.identification: [ { position: 30,  label: 'Accounting Doc' } ]
  key c.belnr as Belnr
```

[A4H] ZPO_CDS_RAP_2031 | [A4H] ZPO_SD_RAP_2031 | [A4H] ZPO_SB_RAP_2031

```
@EndUserText.label: 'Service Definiton for Read-Only Tables'
define service ZPO_SD_RAP_2031 {
  expose ZPO_CDS_RAP_2031;
}
```

Service Binding: ZPO_SB_RAP_2031

General Information

This section describes general information about this service binding

Binding Type: OData V2 - UI

Service Versions

Define service versions associated with the service binding

type filter text

Version	API State	Service Definition
1.0.0	Not Released	ZPO_SD_RAP_2031

Add... | Remove

Service Version Details

View information on selected service version

Local Service Endpoint: Published | Unpublish

Service Information

Service URL: /sap/opu/odata/sap/ZPO_SB_RAP_2031

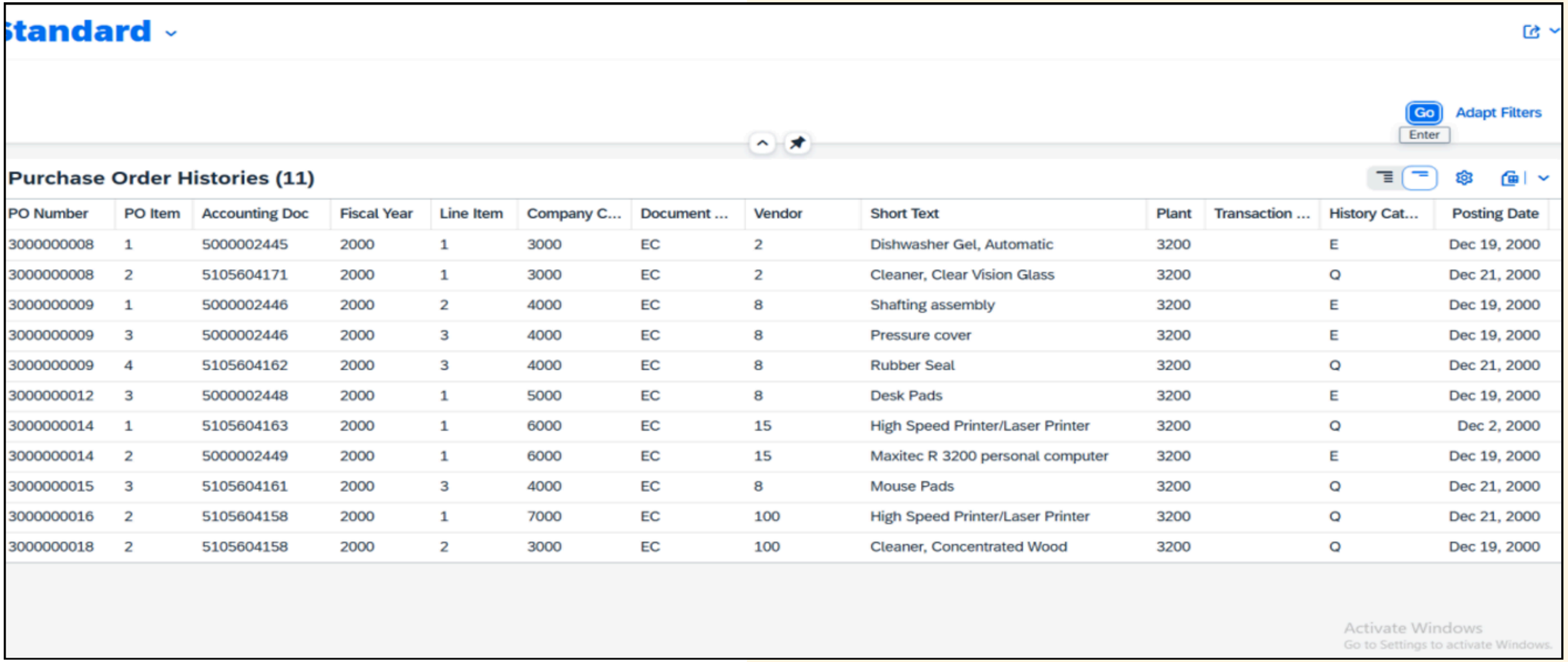
type filter text

Entity Set and Association | Preview...

ZPO_CDS_RAP_2031

RAP READ-ONLY OUTPUT SCREEN

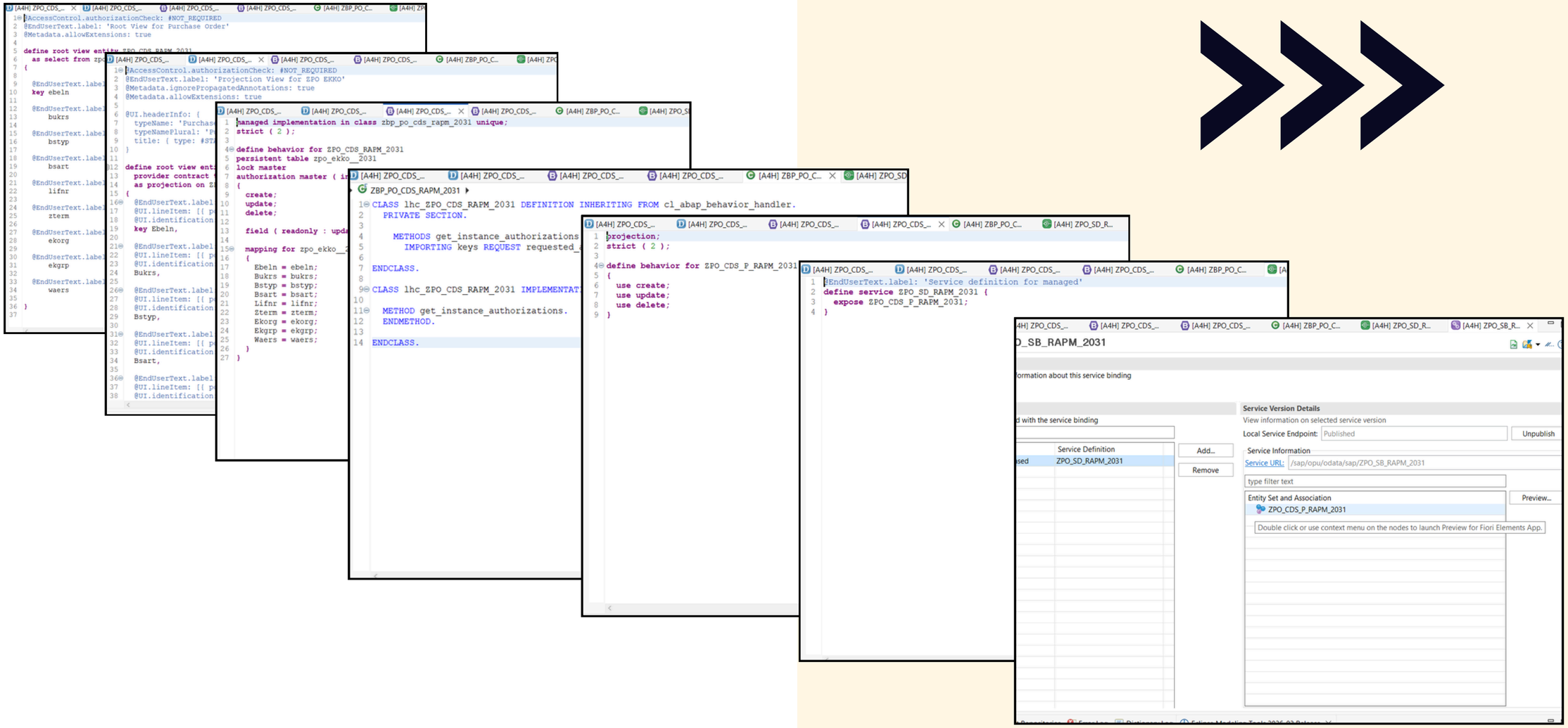
"RAP Read-Only enables secure and efficient data retrieval without allowing any modifications to the underlying data."



PO Number	PO Item	Accounting Doc	Fiscal Year	Line Item	Company C...	Document ...	Vendor	Short Text	Plant	Transaction ...	History Cat...	Posting Date
3000000008	1	5000002445	2000	1	3000	EC	2	Dishwasher Gel, Automatic	3200		E	Dec 19, 2000
3000000008	2	5105604171	2000	1	3000	EC	2	Cleaner, Clear Vision Glass	3200		Q	Dec 21, 2000
3000000009	1	5000002446	2000	2	4000	EC	8	Shafting assembly	3200		E	Dec 19, 2000
3000000009	3	5000002446	2000	3	4000	EC	8	Pressure cover	3200		E	Dec 19, 2000
3000000009	4	5105604162	2000	3	4000	EC	8	Rubber Seal	3200		Q	Dec 21, 2000
3000000012	3	5000002448	2000	1	5000	EC	8	Desk Pads	3200		E	Dec 19, 2000
3000000014	1	5105604163	2000	1	6000	EC	15	High Speed Printer/Laser Printer	3200		Q	Dec 2, 2000
3000000014	2	5000002449	2000	1	6000	EC	15	Maxitec R 3200 personal computer	3200		E	Dec 19, 2000
3000000015	3	5105604161	2000	3	4000	EC	8	Mouse Pads	3200		Q	Dec 21, 2000
3000000016	2	5105604158	2000	1	7000	EC	100	High Speed Printer/Laser Printer	3200		Q	Dec 21, 2000
3000000018	2	5105604158	2000	2	3000	EC	100	Cleaner, Concentrated Wood	3200		Q	Dec 19, 2000

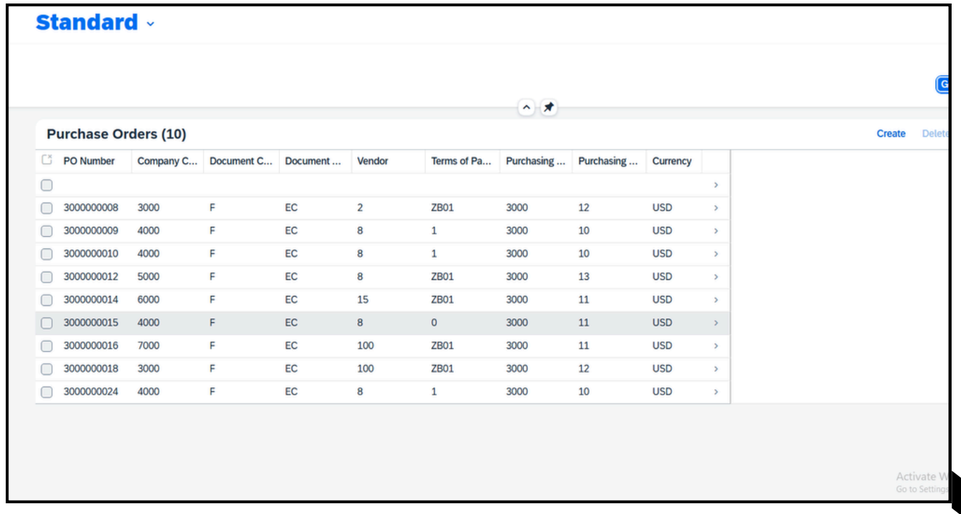
RAP MANAGED INPUT

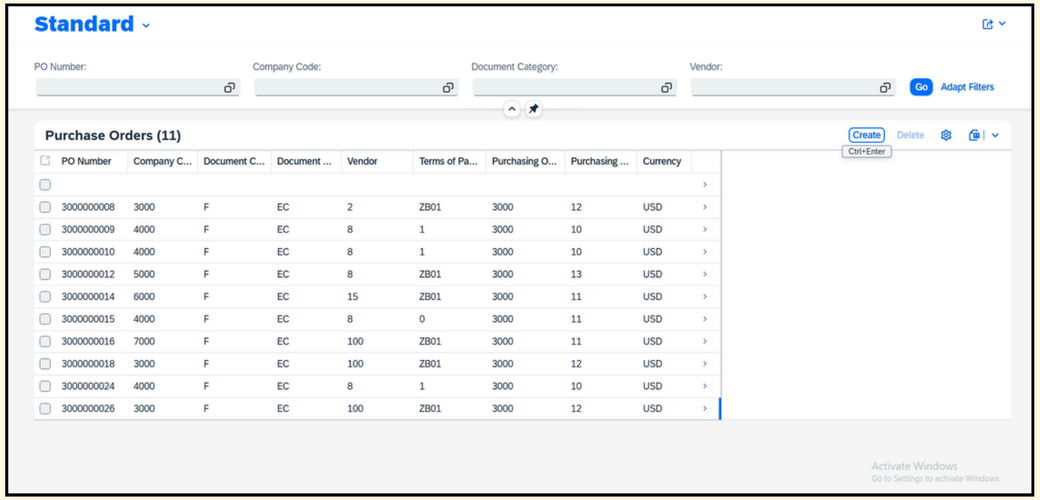
“Manual handling of CRUD operations and transactional logic increases complexity and development effort.”

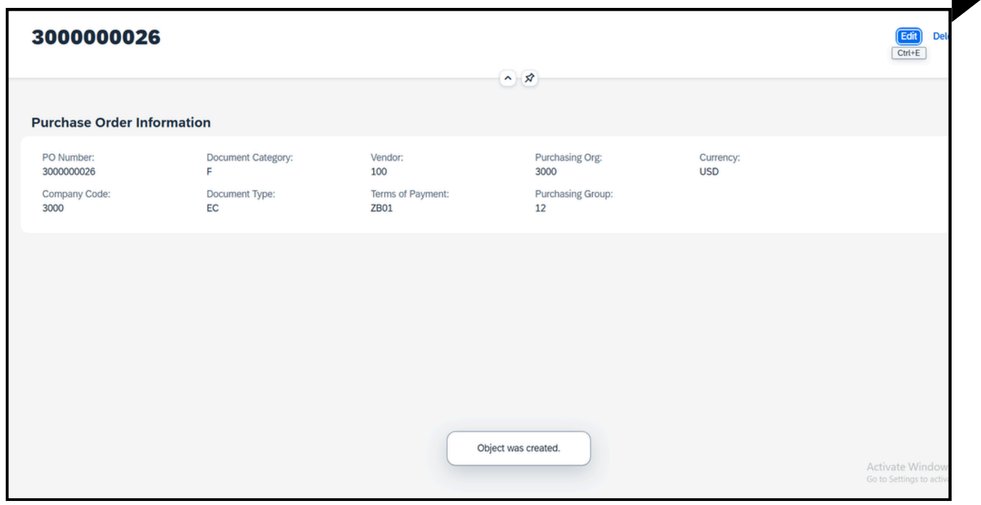


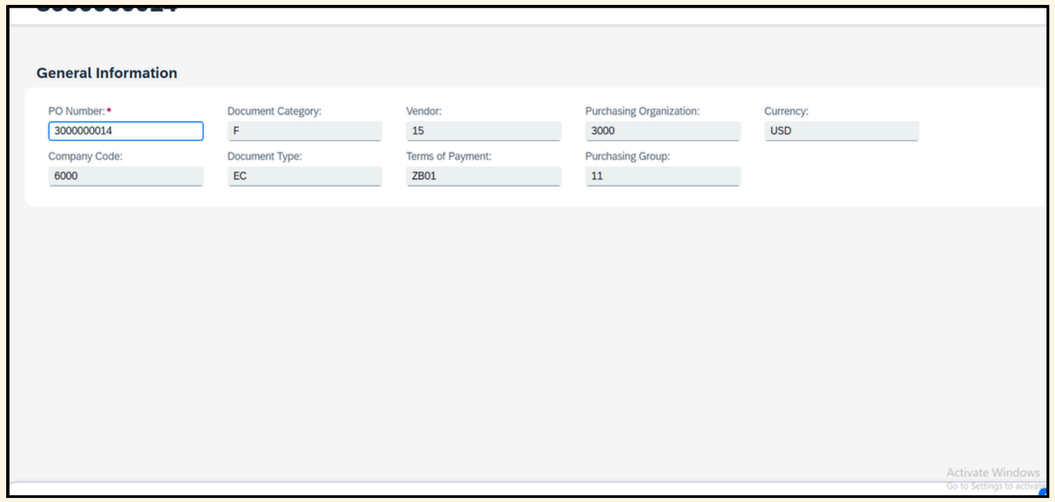
RAP MANAGED OUTPUT

"RAP Managed Scenario enables automatic handling of standard CRUD operations and transactional behavior by the framework."



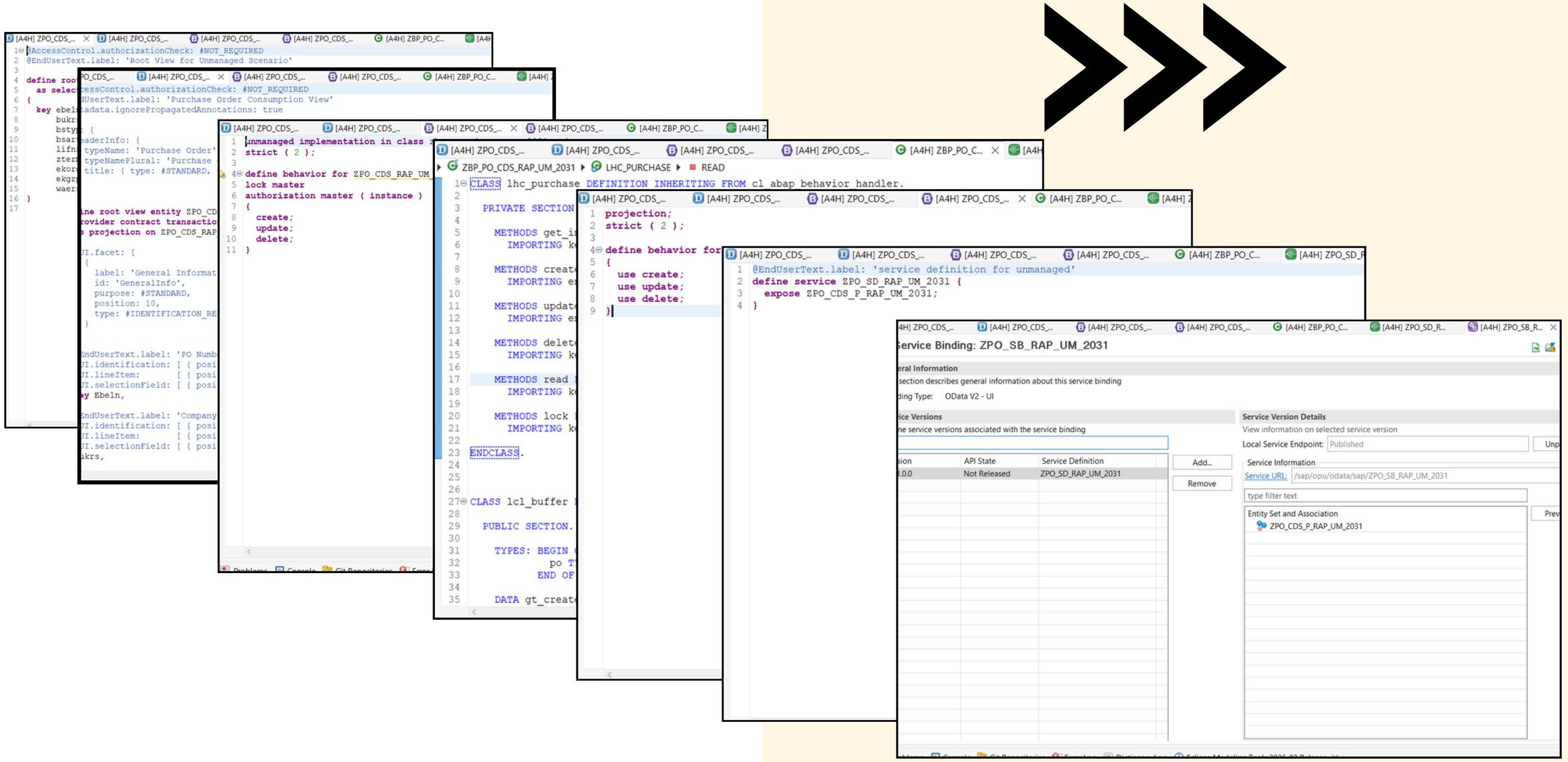






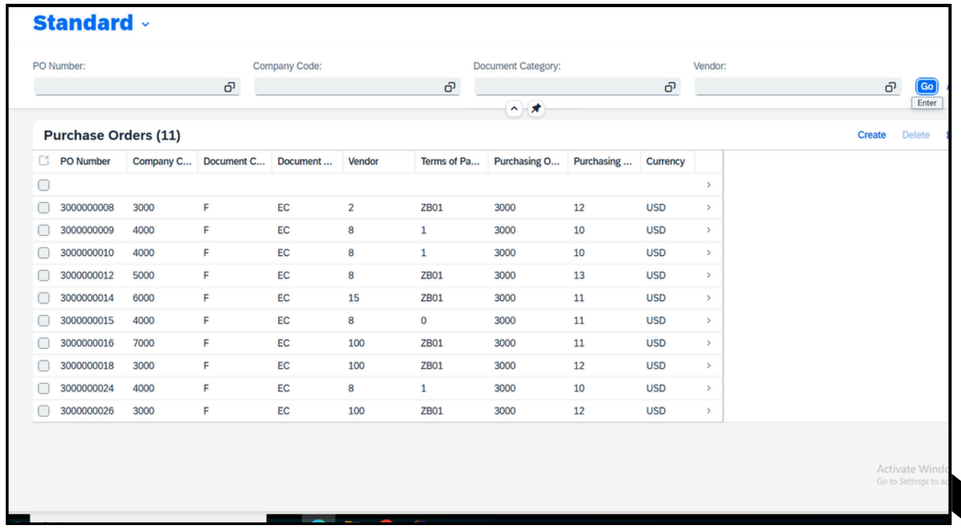
RAP UNMANAGED INPUT

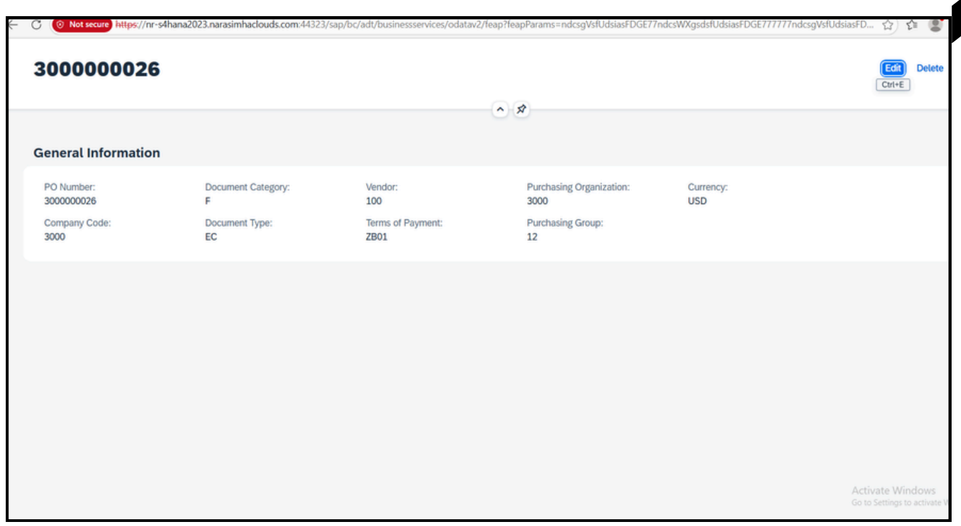
“There is a need for greater control over data operations to implement custom business requirements effectively.”

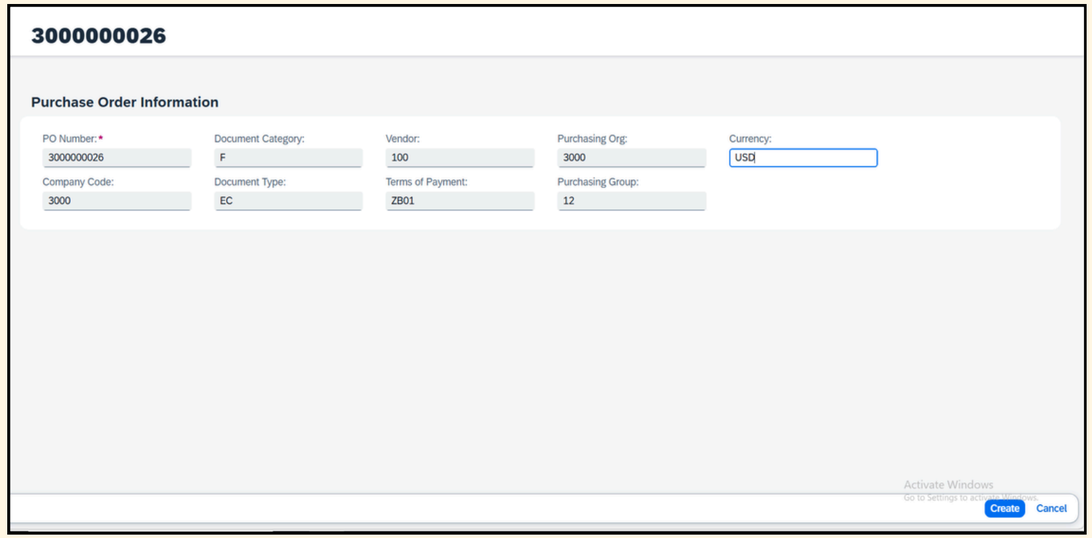


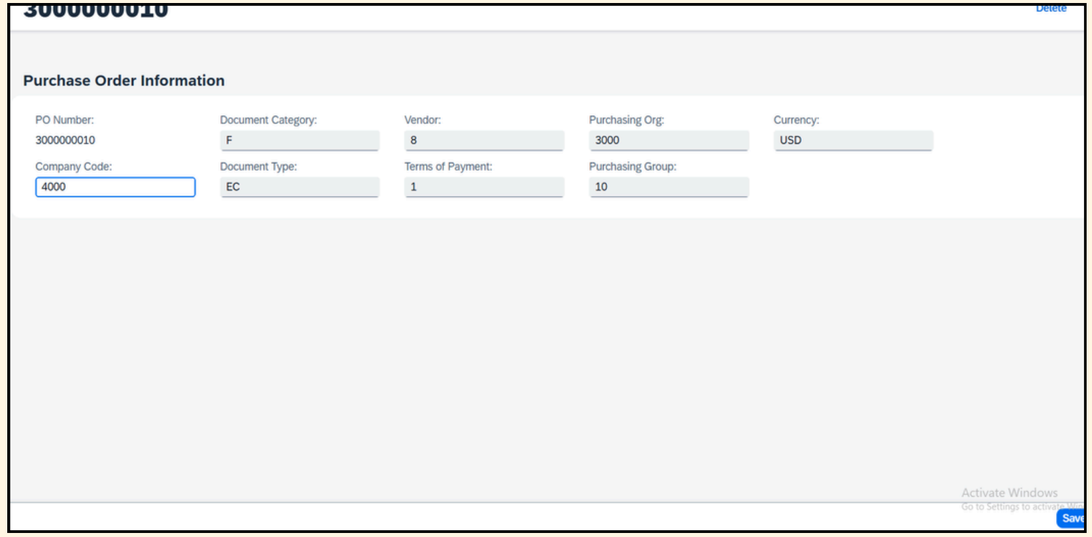
RAP UNMANAGED OUTPUT

“RAP Unmanaged Scenario allows full control over CRUD operations and business logic through custom implementation.”



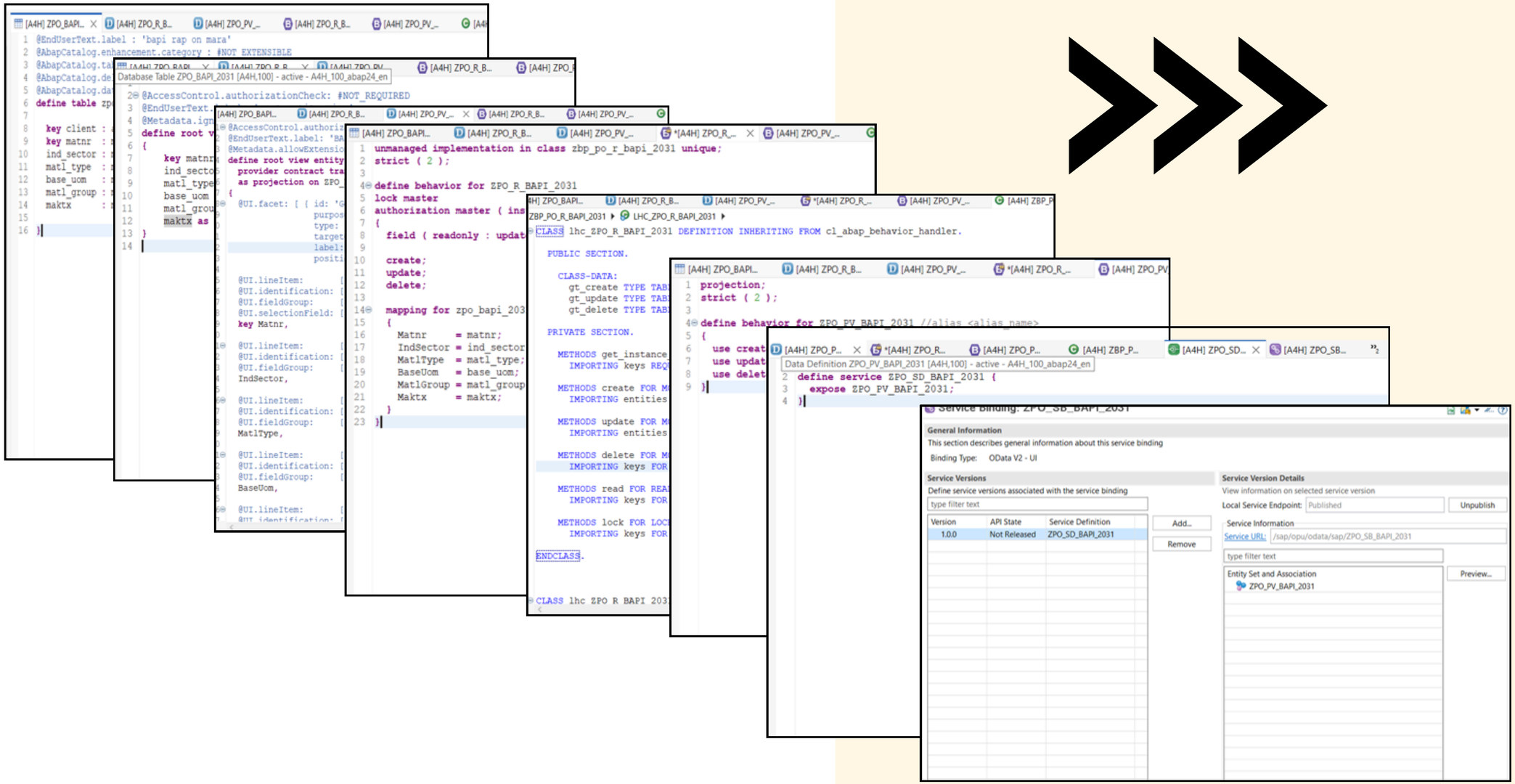






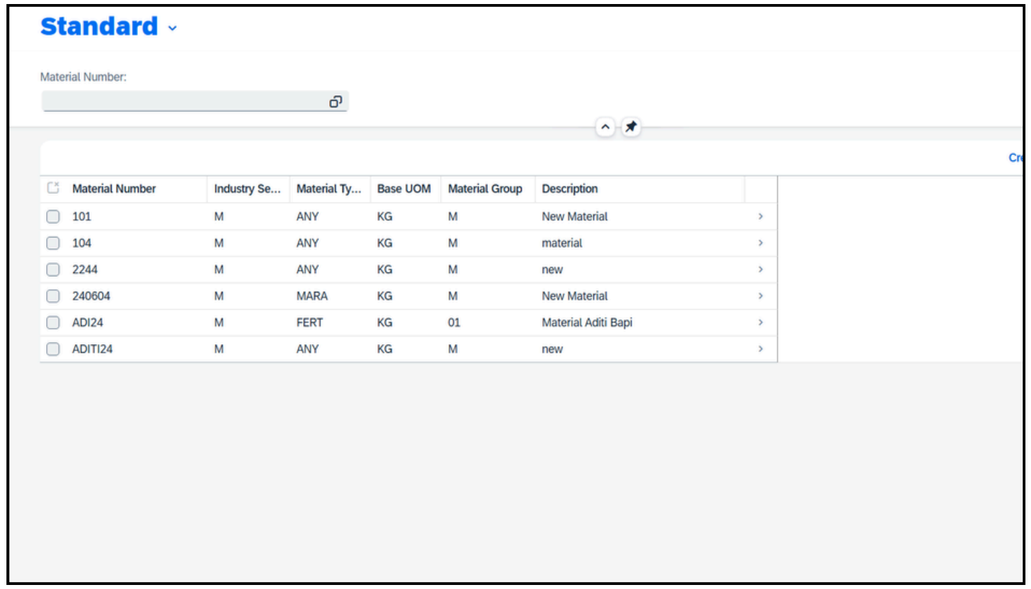
RAP MATERIAL-CREATION BAPI INPUT

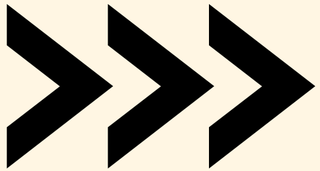
“There is a need to integrate standard business operations within modern frameworks to ensure consistency and reliability.”

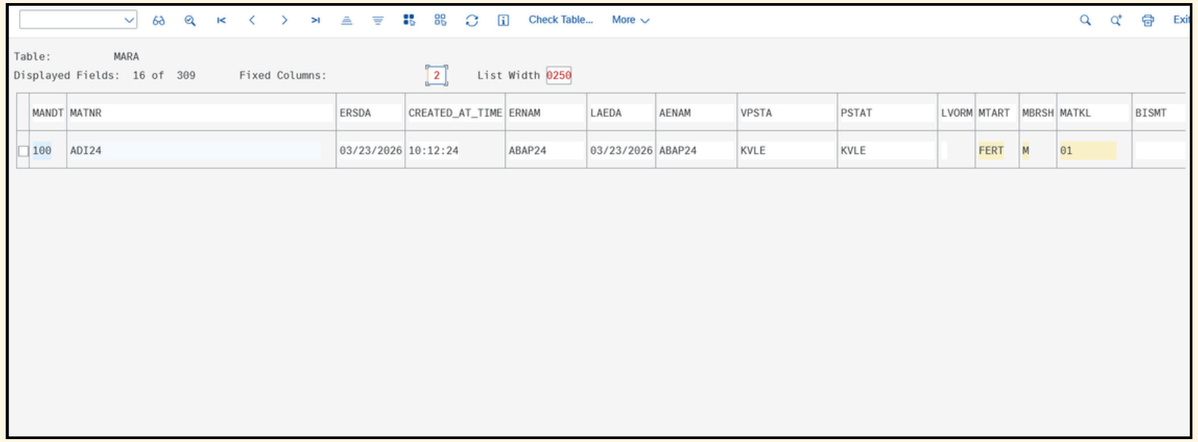


RAP MATERIAL-CREATION BAPI OUTPUT

"Business operations are executed through BAPI within RAP, ensuring structured and consistent processing".

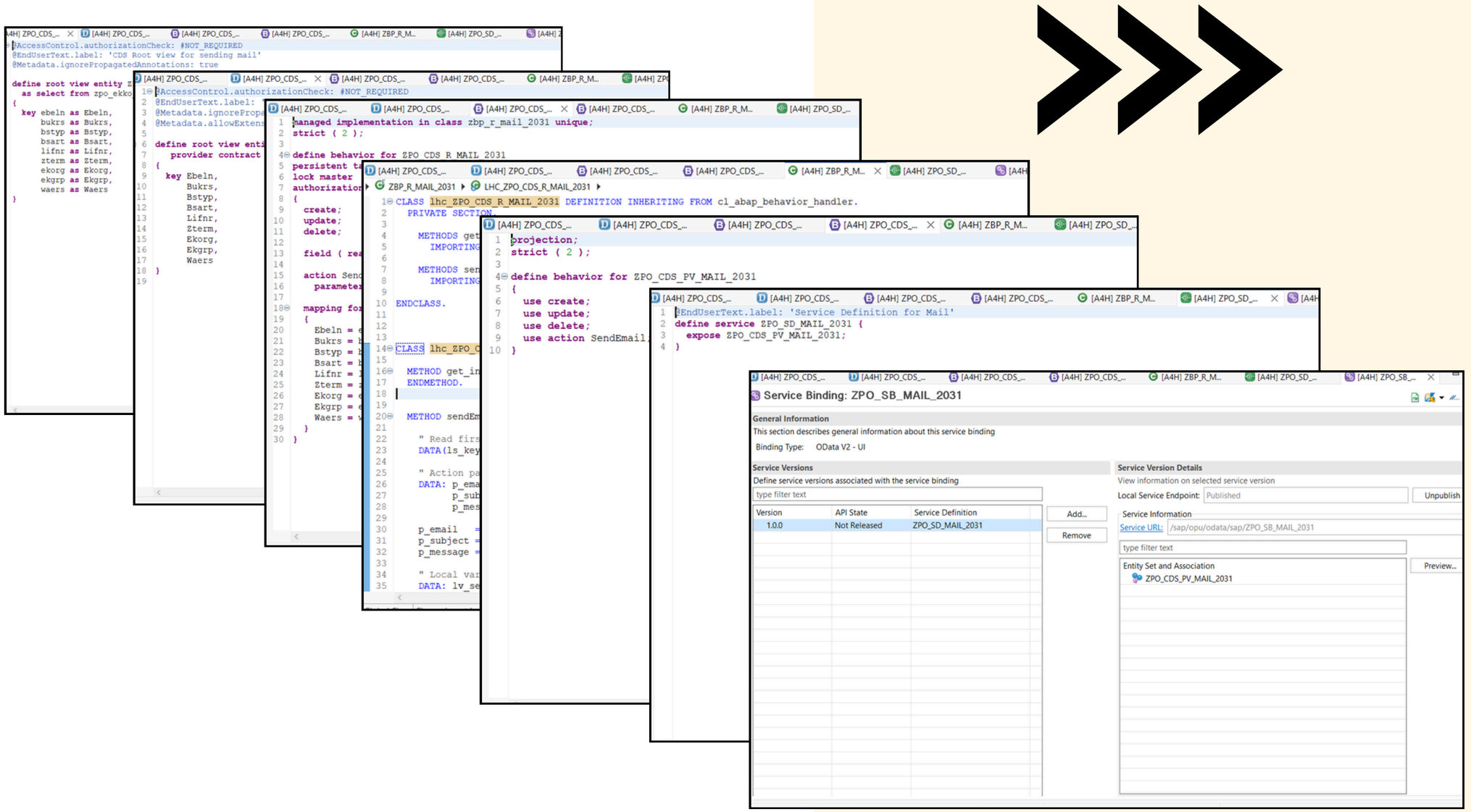






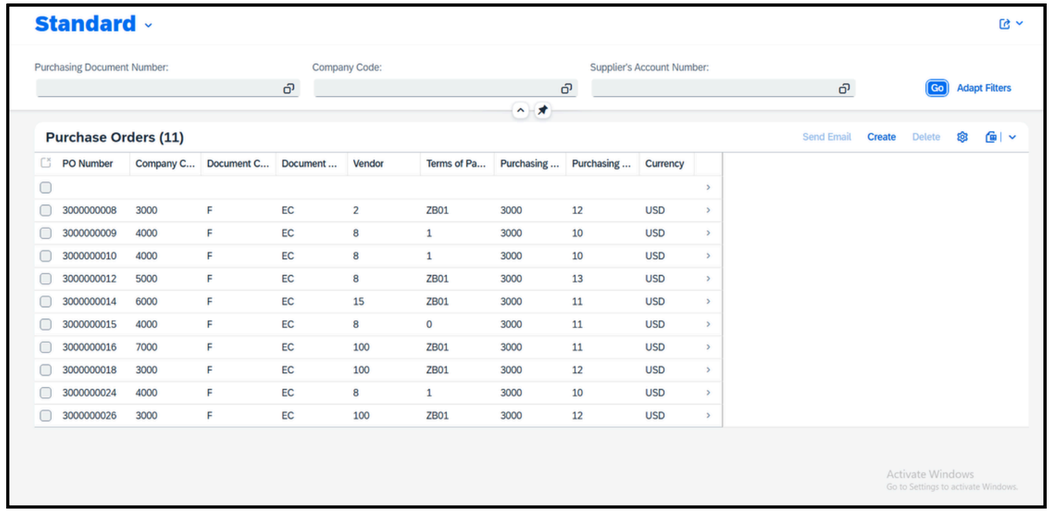
RAP SENDING EMAIL INPUT

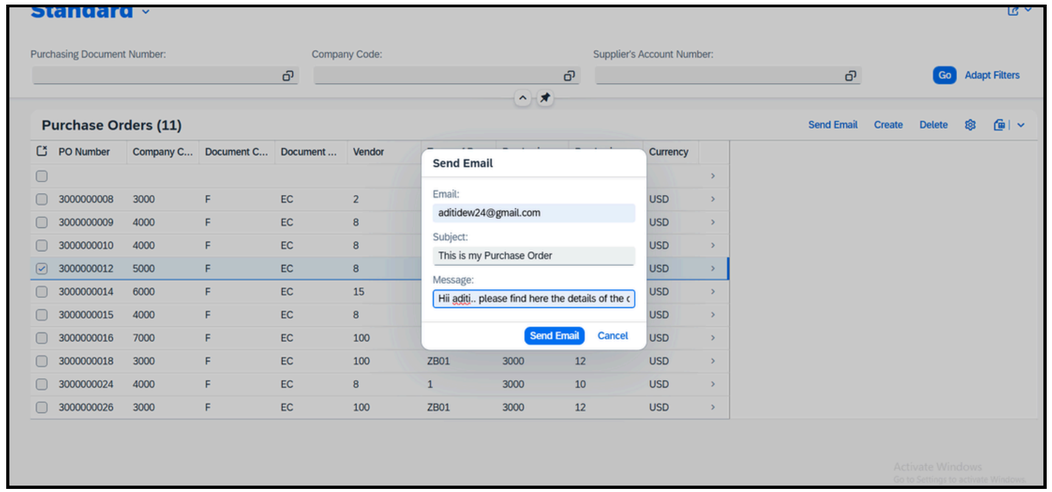
“Systems lack automated communication mechanisms, making it difficult to notify users about important actions and updates.”

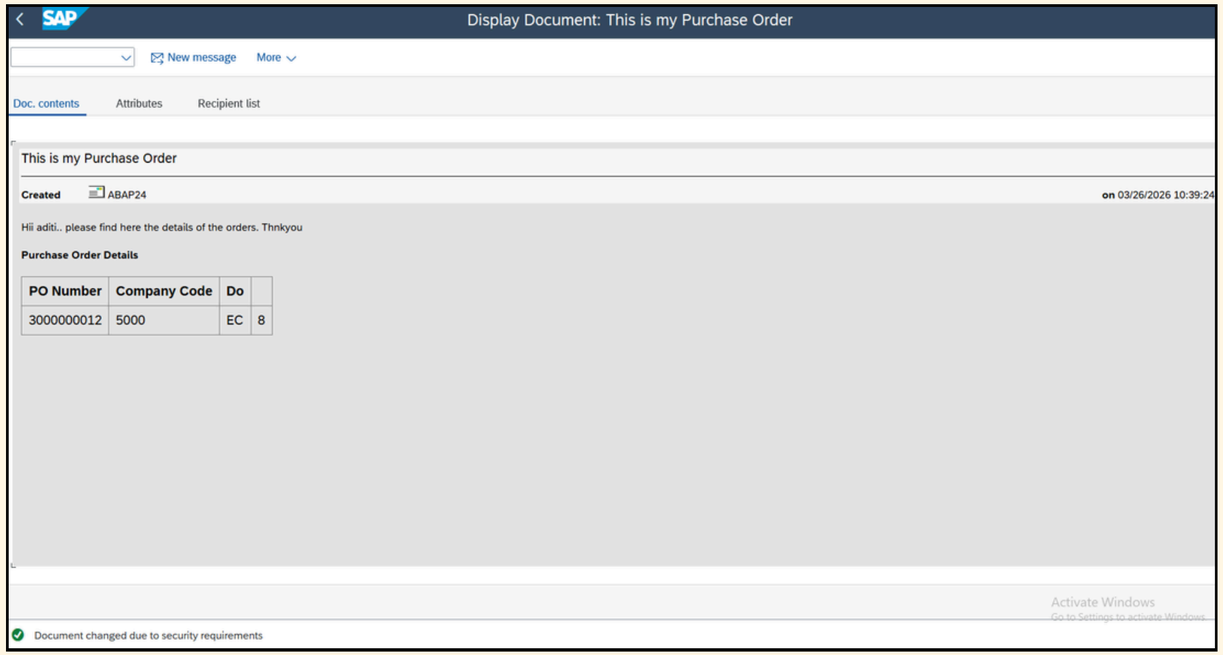


RAP SENDING EMAIL OUTPUT

"The application triggers email notifications through RAP actions, enabling seamless communication."

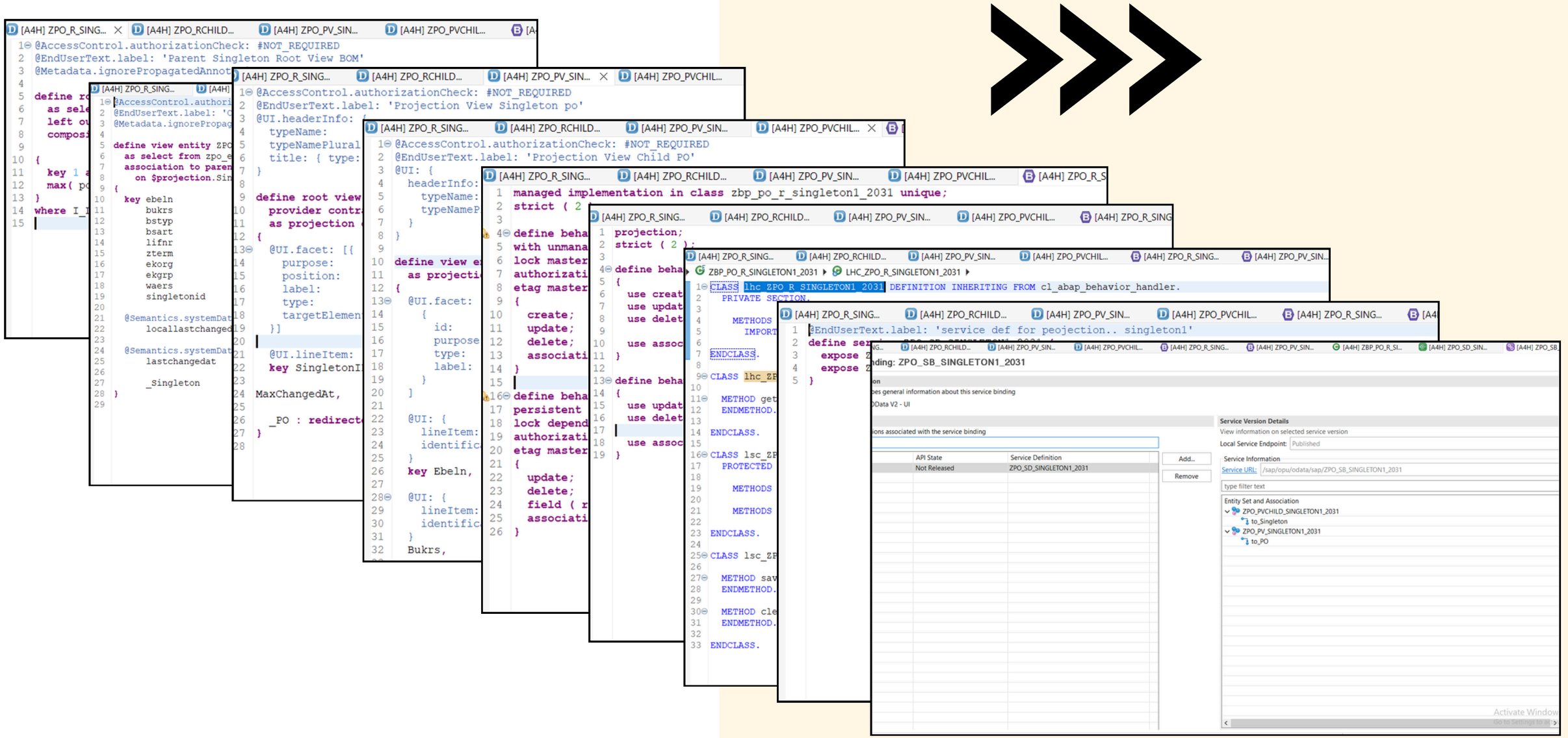






RAP SINGLETON SM30 INPUT

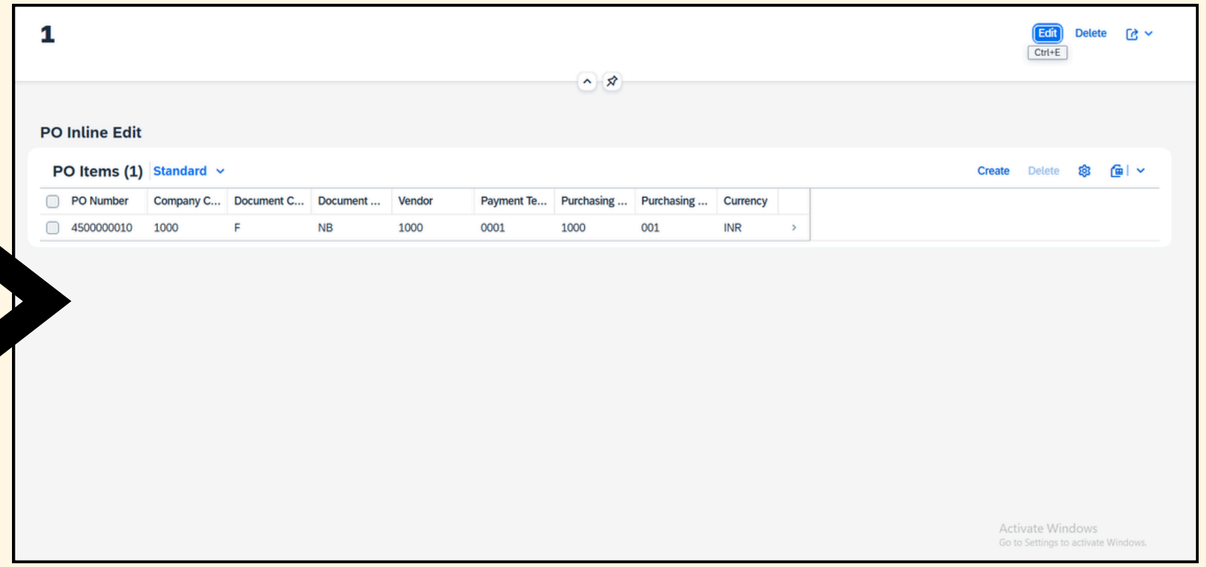
“Managing global data and updating multiple records individually is inefficient and time-consuming.”



RAP SINGLETON OUTPUT

"RAP Singleton enables managing global data with support for multi-inline editing of multiple records efficiently."

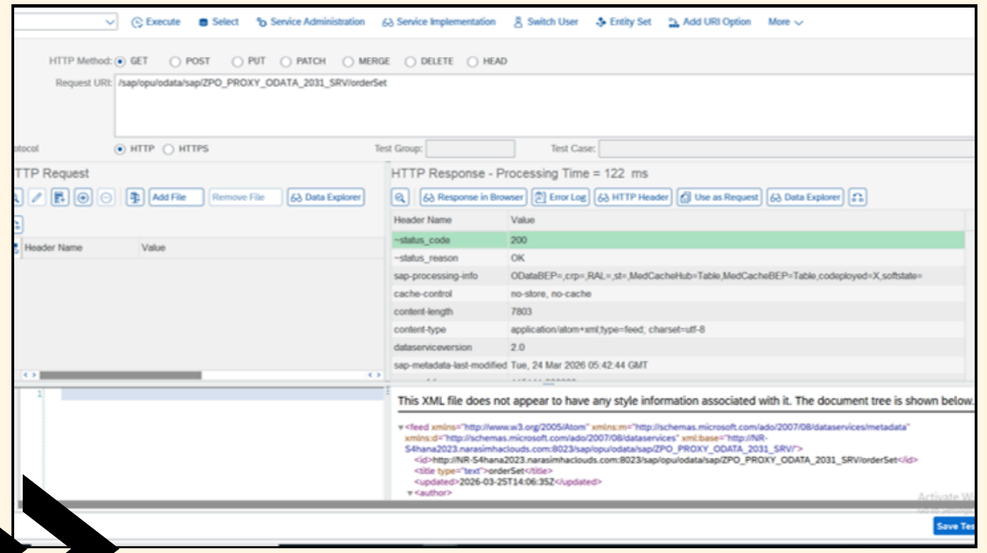


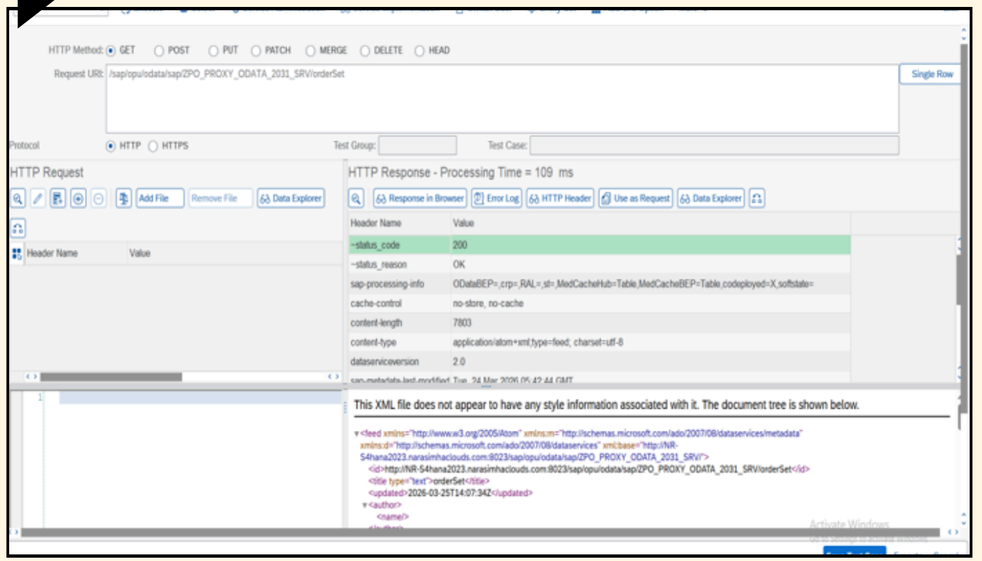


PROXY INPUT/OUTPUT

“Consuming external OData services in ABAP lacks a structured and type-safe approach, making integration complex.”

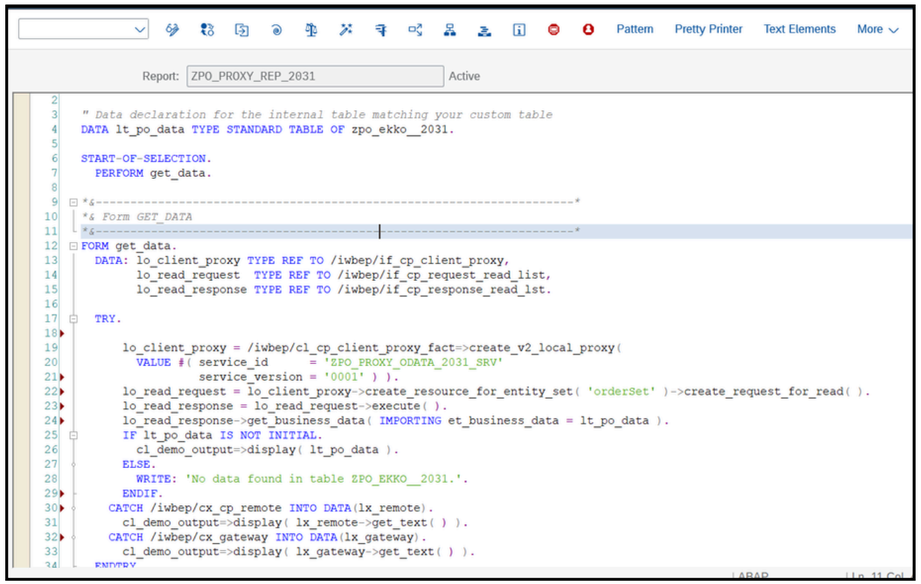
ZABAP24_NET1	Net1 gateway practice for spfli table
ZABAP24_NET2	Netweaver 2 practice without mapping
ZABAP24_NET3	Test Project
ZABAP24_NET4	kna1 crud operation
ZABAP24_NET5	vbak crud operations
ZABAP24_NET6_RAP	ODATA RAP
ZPO_CRUD_ODATA_2031	CRUD Operations in EKKO table
ZPO_ODATA_2031	CRUD Operations in ekko
ZPO_PROXY_ODATA_2031	proxy for case study
Data Model	
Entity Types	
order	
Associations	
Entity Sets	
orderSet	
Function Imports	
GetPO	
Function Import Parameters	
Ebeln	
Service Implementation	
orderSet	





PROXY INPUT/OUTPUT

“Consuming external OData services in ABAP lacks a structured and type-safe approach, making integration complex.”



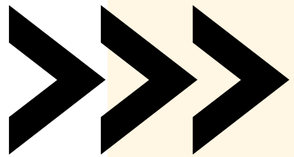

```
" Data declaration for the internal table matching your custom table
DATA lt_po_data TYPE STANDARD TABLE OF zpo_ekko__2031.

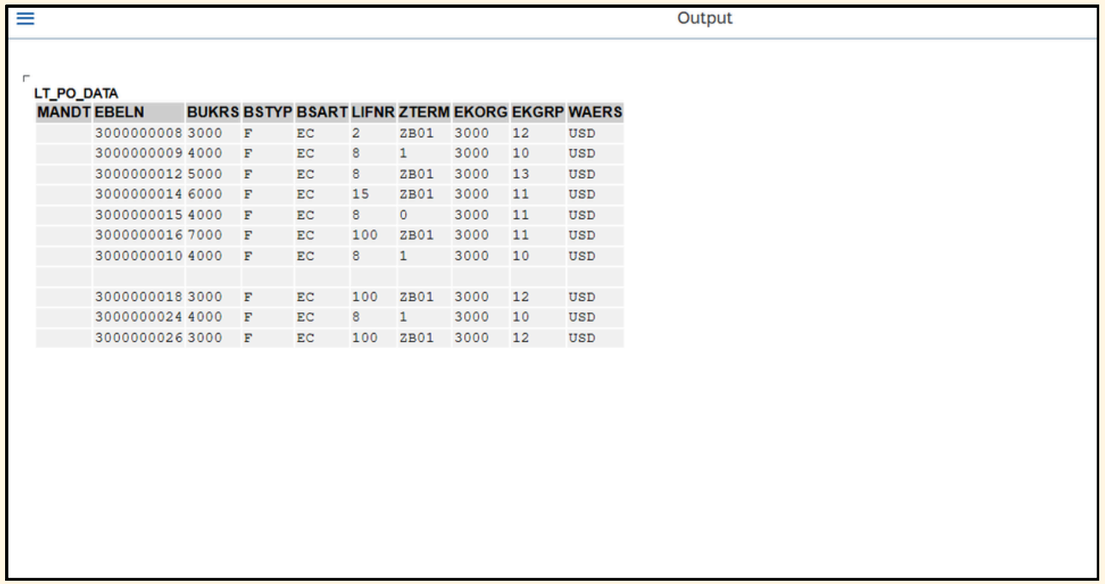
START-OF-SELECTION.
  PERFORM get_data.

*&---------------------------------------------------------------------*
*& Form GET_DATA
*&---------------------------------------------------------------------*
FORM get_data.
  DATA: lo_client_proxy TYPE REF TO /iwbep/if_cp_client_proxy,
        lo_read_request  TYPE REF TO /iwbep/if_cp_request_read_list,
        lo_read_response TYPE REF TO /iwbep/if_cp_response_read_lst.

  TRY.

      lo_client_proxy = /iwbep/cl_cp_client_proxy_fact=>create_v2_local_proxy(
        VALUE #( service_id      = 'ZPO_PROXY_ODATA_2031_SRV'
                 service_version = '0001' ) ).
      lo_read_request = lo_client_proxy->create_resource_for_entity_set( 'orderSet' )->create_request_for_read( ).
      lo_read_response = lo_read_request->execute( ).
      lo_read_response->get_business_data( IMPORTING et_business_data = lt_po_data ).
      IF lt_po_data IS NOT INITIAL.
        cl_demo_output=>display( lt_po_data ).
      ELSE.
        WRITE: 'No data found in table ZPO_EKKO__2031.'.
      ENDIF.
    CATCH /iwbep/cx_cp_remote INTO DATA(lx_remote).
      cl_demo_output=>display( lx_remote->get_text( ) ).
    CATCH /iwbep/cx_gateway INTO DATA(lx_gateway).
      cl_demo_output=>display( lx_gateway->get_text( ) ).
  ENDTRY.
```





LT_PO_DATA

MANDT	EBELN	BUKRS	BSTYP	BSART	LIFNR	ZTERM	EKORG	EKGRP	WAERS
	3000000008	3000	F	EC	2	ZB01	3000	12	USD
	3000000009	4000	F	EC	8	1	3000	10	USD
	3000000012	5000	F	EC	8	ZB01	3000	13	USD
	3000000014	6000	F	EC	15	ZB01	3000	11	USD
	3000000015	4000	F	EC	8	0	3000	11	USD
	3000000016	7000	F	EC	100	ZB01	3000	11	USD
	3000000010	4000	F	EC	8	1	3000	10	USD
	3000000018	3000	F	EC	100	ZB01	3000	12	USD
	3000000024	4000	F	EC	8	1	3000	10	USD
	3000000026	3000	F	EC	100	ZB01	3000	12	USD



THANK YOU!!

I’d be happy to take any questions.